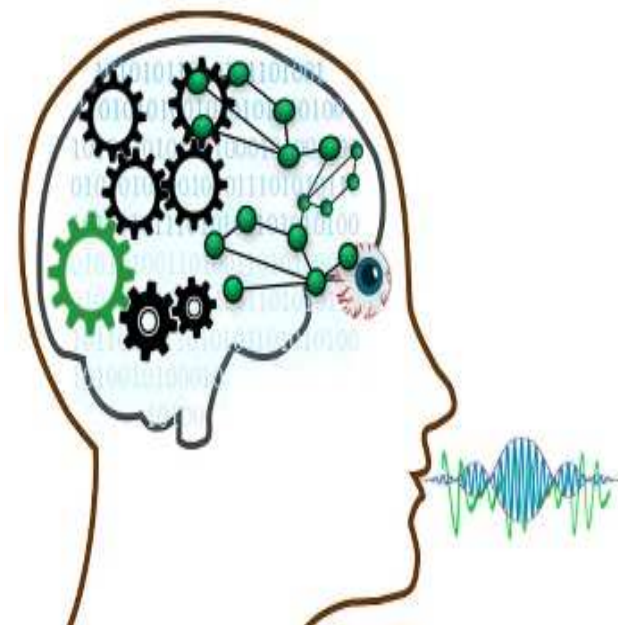
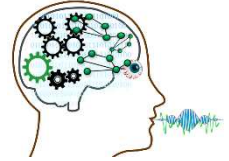




Vision AI Introduction

Software Engineering and Data Science
SEIS 766: Vision AI
Dr. Chih Lai

Conventional CNN, Whole Image Classification



Dogs



Plants



Bikes



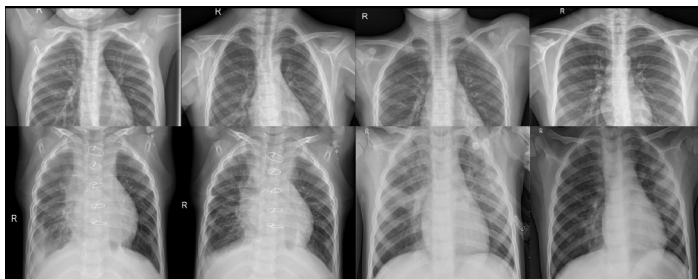
Muffin or Chihuahua



StateFarm™ kaggle



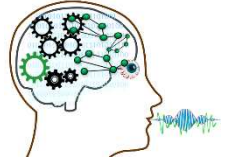
Healthy vs Pneumonia



COVID-19



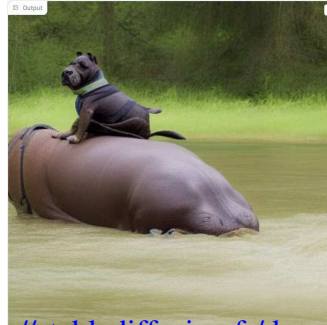
Just Use Those Famous Popular Tools???



- OpenAI, Sora, DALL-E
- Google Veo 3, Nano Banana
- Meta SeeDream, DINOv2,
- Stable Diffusion

TEXT prompt→

*Dog ridding on
the back of a hippo*



<https://stablediffusion.fr/demo>

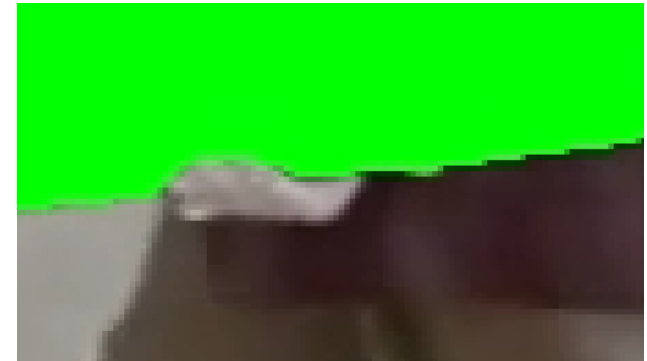
Stable Diffusion 2.1 Demo



<https://huggingface.co/spaces/stabilityai/stable-diffusion>



https://aidemos.meta.com/segment-anything/gallery/?template_id=813317788097025

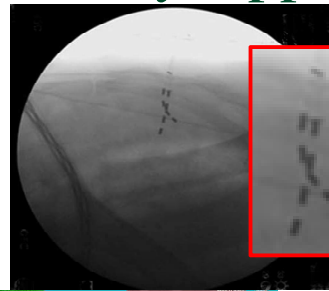
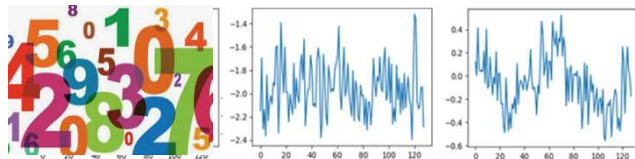


“Tools help you finish today’s task, but the skill you learn help you build a career.”

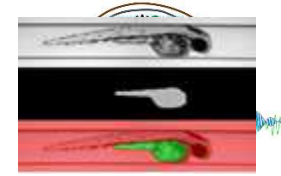
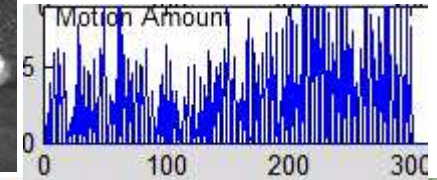
“Tools change; fundamentals transfer.”

Precision Is Everything In Many Apps

Numeric Data



Frq_parkinsonian tremors.avi

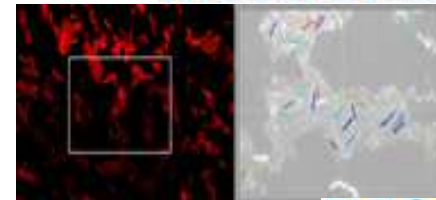
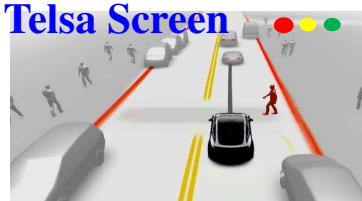


Calorie
\$\$\$



<https://youtu.be/hxczTe9U8jY>

Telsa Screen



ECOLAB



Auto Property Listings
Prevent Fraud
Improve Query / suggest



<https://www.winnowsolutions.com/>

<https://www.leanpath.com/>

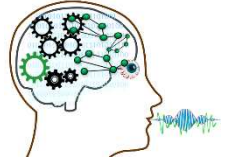
1:15



<https://www.youtube.com/watch?v=IZj7YVa9YLI>

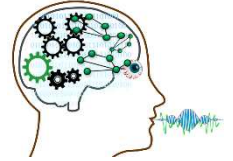


Outline

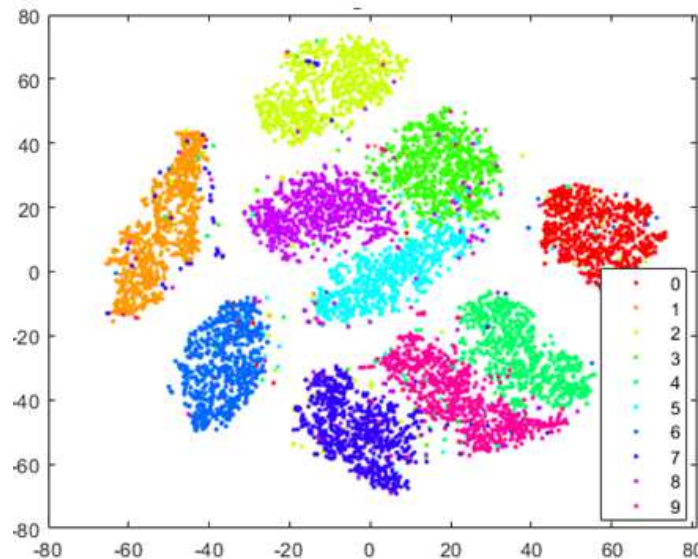


- ➔ **Z** / Visual Embedding
 - Object Allocations by Pixel Classification
 - “**COLORS**” to VAI as “Words” to Conv. AI
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 - Quality Eval of Generation, *Motion*, *Image Processing* ➔ *image light, resolution*

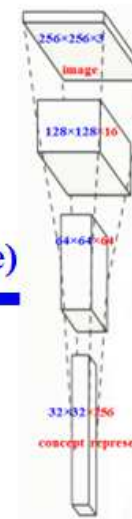
Visual Embedding / Z



- **Visual Embedding / Z**, hidden representation of images like word/sentence embedding
- We know that well enough... any (new) applications?? other than for visual??
- **Many** (most) of VAI methods based on Z



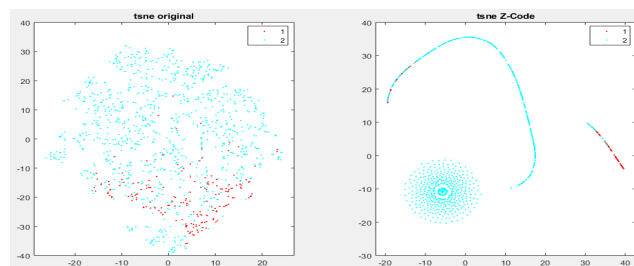
Ordinary
CNN



MNIST dataset



tsne(Z_Code)

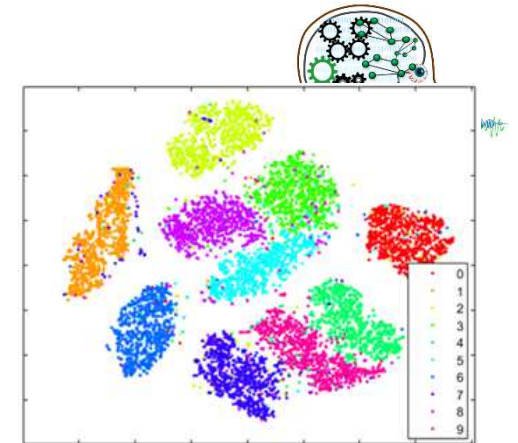
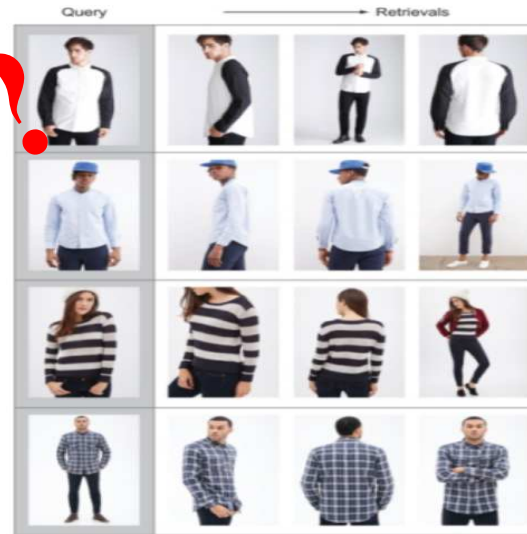


Z-Code

	X														
#	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Y
1	222	31.18919	40.34234	35.57909	8.883917	0.968325	-80.1137	222	1	16.81247	0.816176	0.578125	78.591	0	
2	73	29.49315	271.3973	15.5172	6.40749	0.910764	76.04295	73	1	9.640876	0.858824	0.608333	39.217	0	
3	256	58.81641	289.9414	37.22601	9.863895	0.964256	85.32474	256	1	18.05407	0.752941	0.562637	89.111	0	
4	126	71.02381	477.4127	13.11298	12.79067	0.220351	63.52348	126	1	12.66602	0.881119	0.646154	43.832	0	
5	225	90.80889	541.9467	44.46311	7.858879	0.984256	-52.875	225	1	16.92569	0.728155	0.252525	90.072	7	
6	233	91.40773	279.6223	40.03947	9.256891	0.972908	80.28062	233	1	17.22396	0.732704	0.524775	88.623	0	
7	288	109.7118	187.7882	42.20114	9.184247	0.976031	-0.63687	288	1	19.14923	0.842105	0.738462	87.192	0	
8	260	102.3808	246.4115	35.06891	12.8204	0.930781	77.95412	263	-1	18.19457	0.638821	0.46595	99.788	0	
9	231	106.0563	537.2078	39.62008	8.049715	0.979143	-48.1493	232	0	17.14988	0.868421	0.316872	80.958	0	
10	100	100.94	167.05	19.943	8.591027	0.902457	-88.9441	100	1	11.28379	0.757576	0.584795	48.141	0	
11	74	100.027	505.8919	14.15669	6.873529	0.874219	-82.0348	74	1	9.706685	0.91358	0.813187	31.534	0	
12	102	103.2451	205.4608	13.39616	11.04129	0.56628	-88.6751	103	0	11.39607	0.842975	0.713287	43.435	0	
13	424	118.1226	230.9835	66.00121	12.65948	0.981433	89.0279	425	0	23.23475	0.542894	0.422732	200.18	0	
14	194	136.4742	566.4278	38.25739	7.172156	0.98227	-37.5776	194	1	15.7165	0.788618	0.281159	80.85	0	
15	208	136.4231	165.4952	32.34916	9.873596	0.952282	67.69716	208	1	16.27372	0.8125	0.481481	70.892	0	
16	140	133.154	713.3178	44.47461	6.454235	0.884376	88.61055	140	1	14.75666	0.601489	0.578455	97.054	0	

More Apps w/ Visual Embedding Z

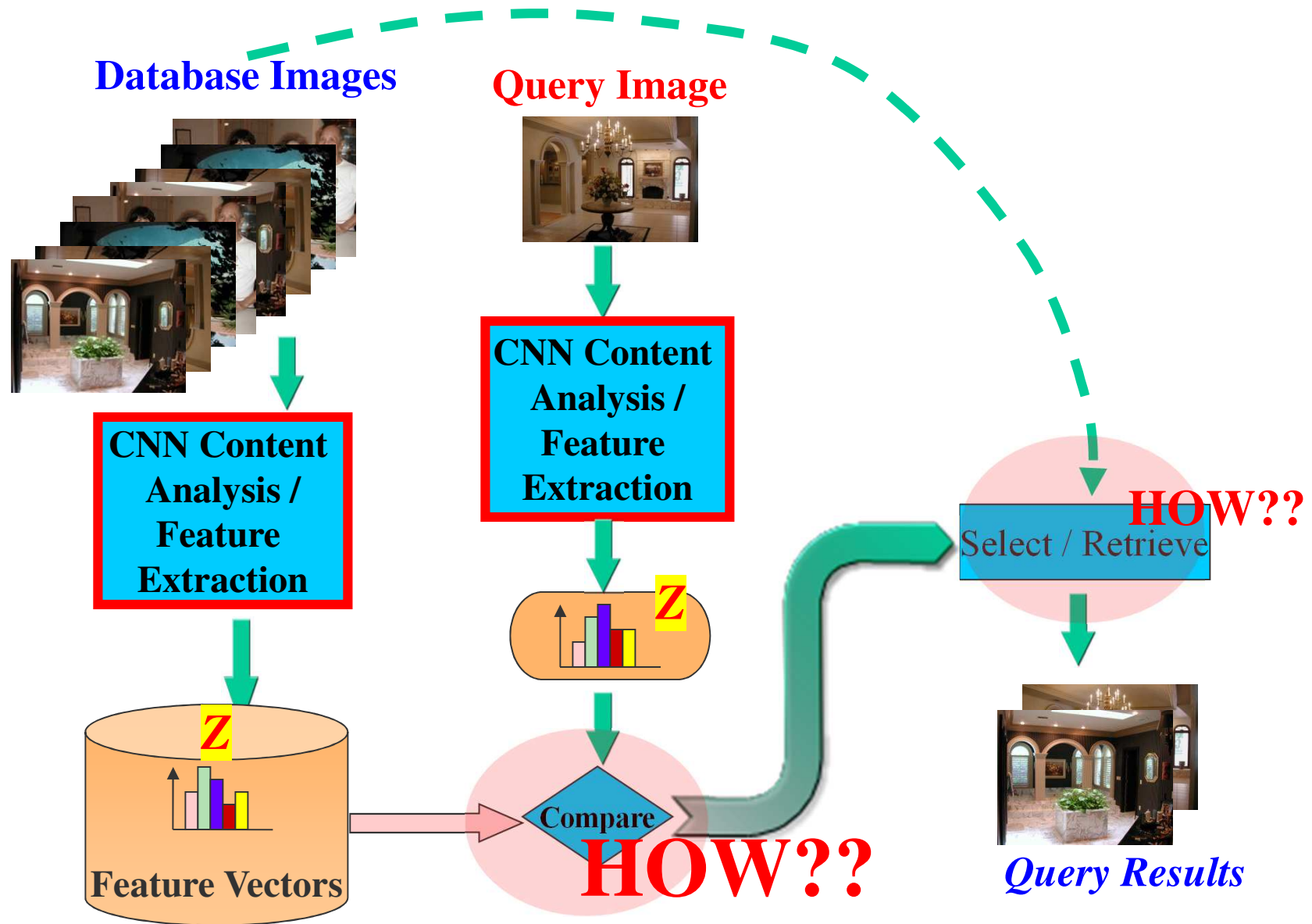
- Image summarization / retrieval using Z ... HOW??
 - Visual recommender / virtual try-on



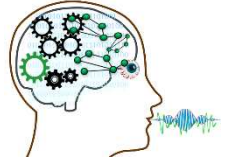
- Integrating image Z with text Z + ...
 - “I want clothes with strips of red & blue”
 - **Multi-Modal Model**
- **Walking** in latent space Z
- **Almost** all other VAI topics...



Retrieval in Visual Recommender Through Visual Embedding **Z**



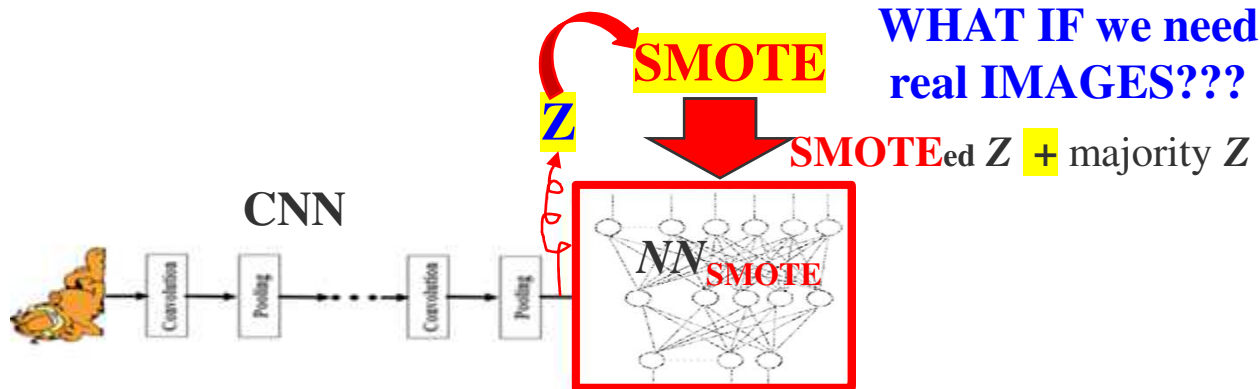
Apply **SMOTE** to Unbalanced Image Dataset????



- **Dynamic** augmentation → unbalanced classes are still unbalanced
- Apply **SMOTE** to oversample **images** in minority classes?? **NOT** directly...

- 1. Instead, take the **Z**-code output from certain layer (i.e. flatten or late conv.)
- 2. Applied **SMOTE** to oversample **Z**-code from minority classes as “**SMOTE_{ed} Z**”
- 3. Train new NN_{SMOTE} on minority “**SMOTE_{ed} Z**” + majority **Z** using just dense layers

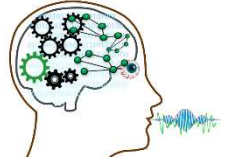
- For test images, get **Z** from CNN first, then input **Z** to NN_{SMOTE} for classification



**ID COVID Patients w/ CNN
SEIS 764 AI, 25F**

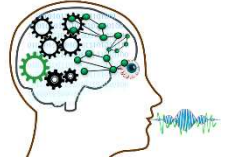
David Braun
Erik Knutsen
Karryn Leake
Clayton Madsen
Dante Razo

Outline

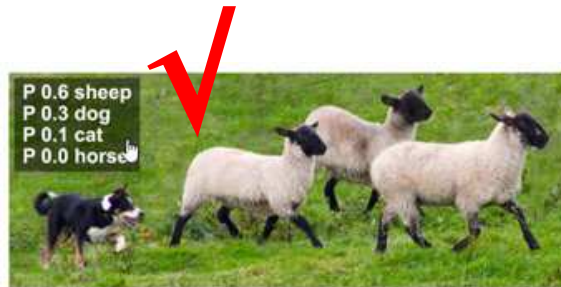


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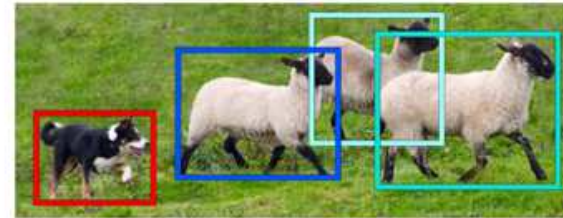
Object Detection / Identification



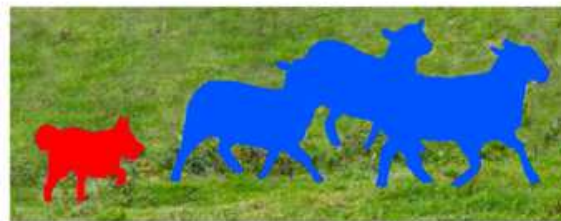
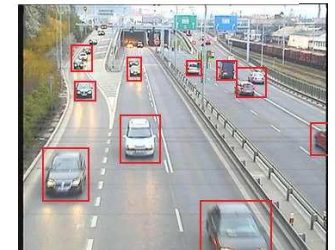
- More than whole image classification & many important applications, like



whole image classification



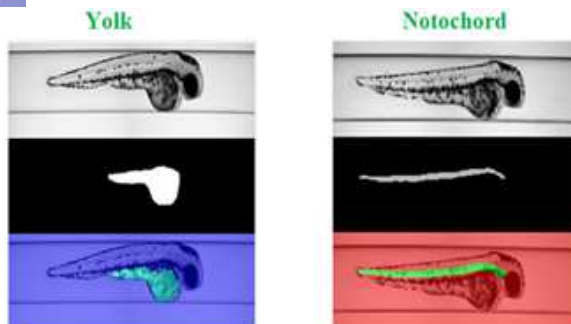
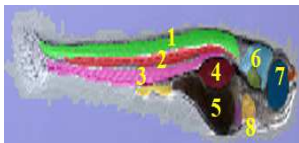
Object Detection



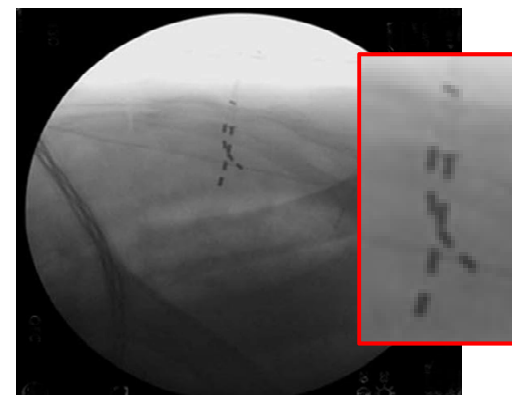
Semantic Segmentation



Instance Segmentation



Objects all look *different*



Objects all look *similar*

Object Detection → Semantic Segmentation → *Instance Segmentation*

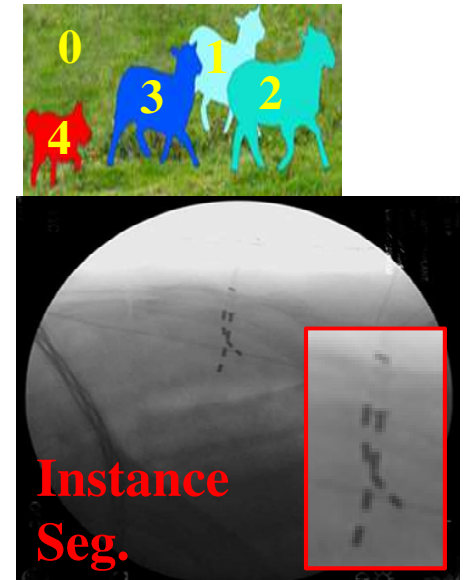


Region Classification

Objects all look *different*

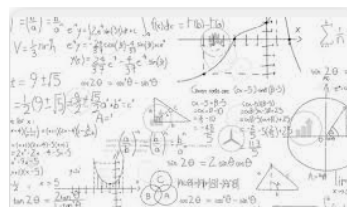


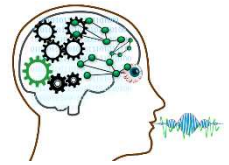
Objects all look *similar*



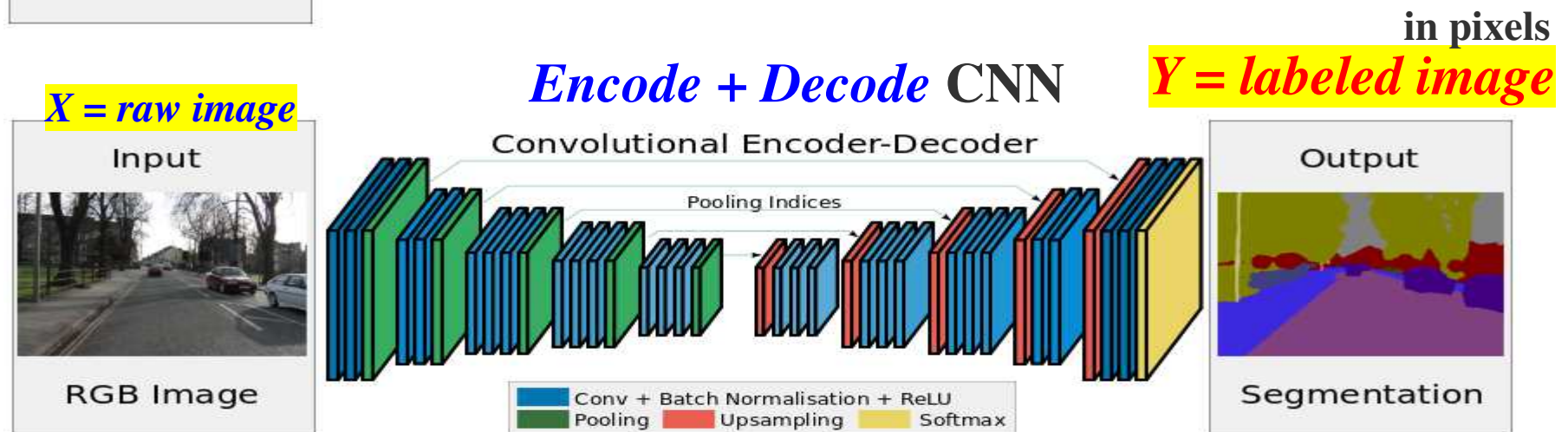
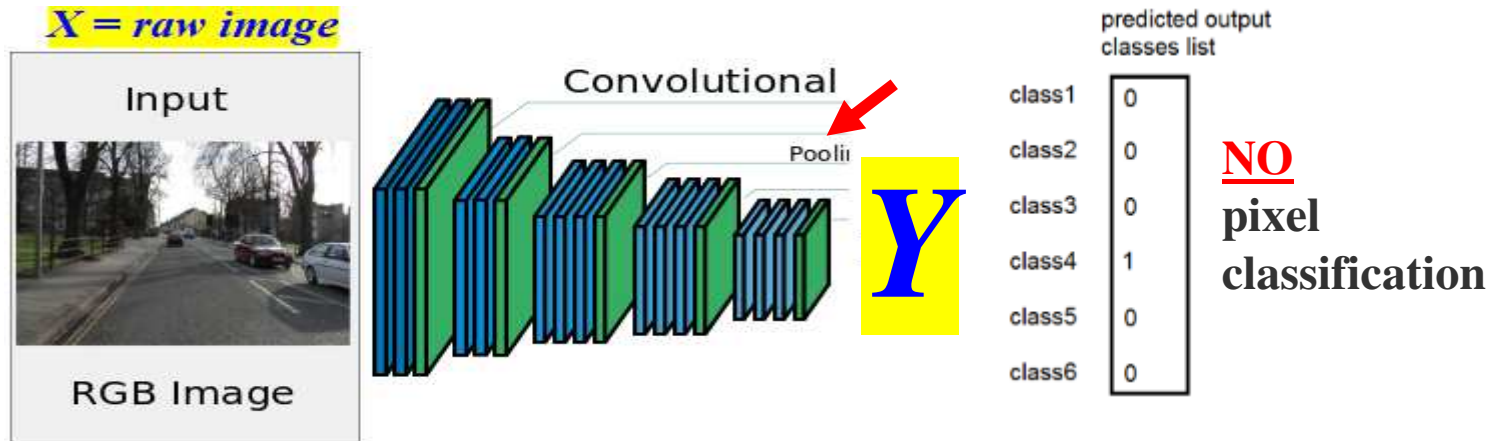
Why show this bad result?? **Pixel Classification**

- Results have many broken artifacts / classes (big / small) all over the places
 - Why showing this bad result?? **Reality!!!** Big deal?
 - Misconception:** judging project time based on “*are images easy for human eyes??*”
 - Require rigid image processing (color / morphological ops) after deep learning



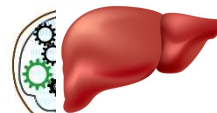


Pixel Classification using CNN Classification??



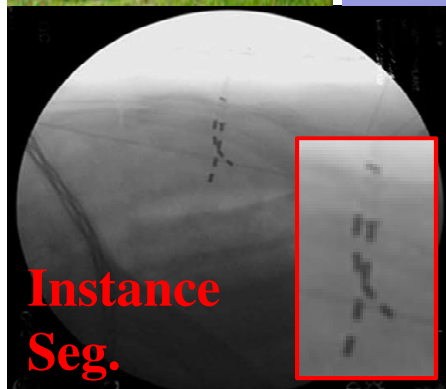
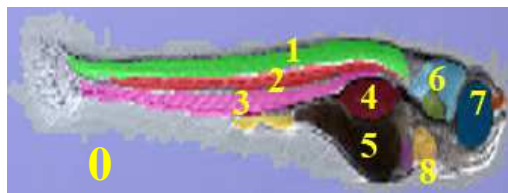
- It requires **pixel classification** of very different **Y s**
- How do we get label images **Y** in pixels?? Who creates them??
- Augmenting **BOTH** **Raw** Images and **Label** Images





“COLORs” Is The Foundations of VAI

■ Problems in Pre-Processing / Augmentation



resizing

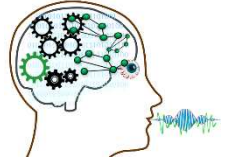


■ Problems in Post-Processing



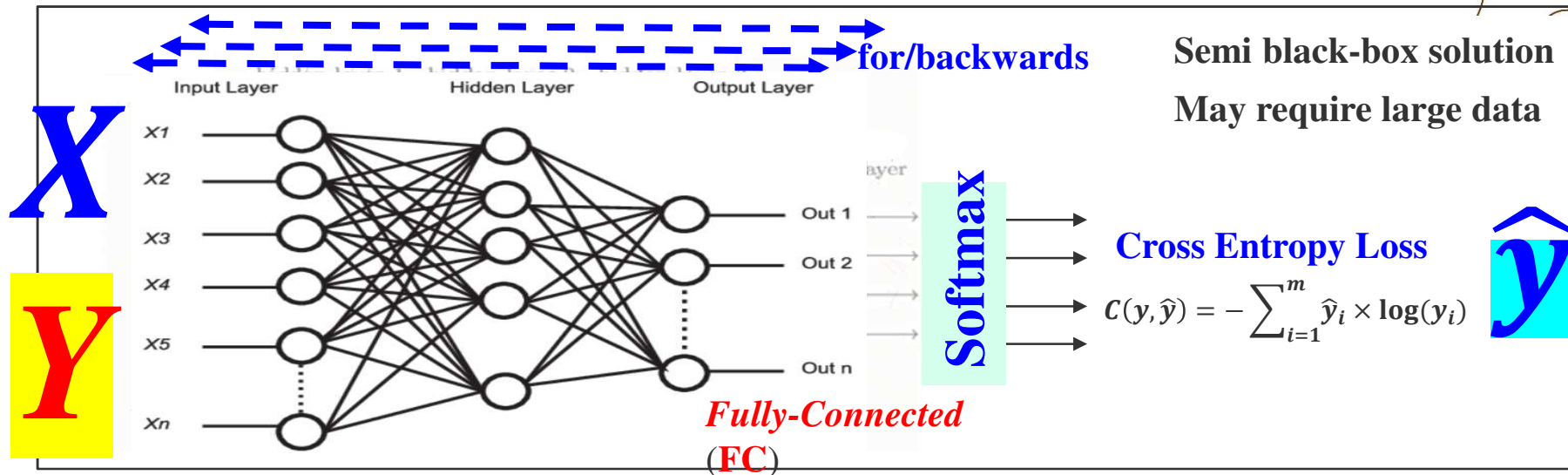
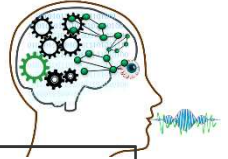
“COLORs” to VAI
as
“Words” to Conv. AI

Outline

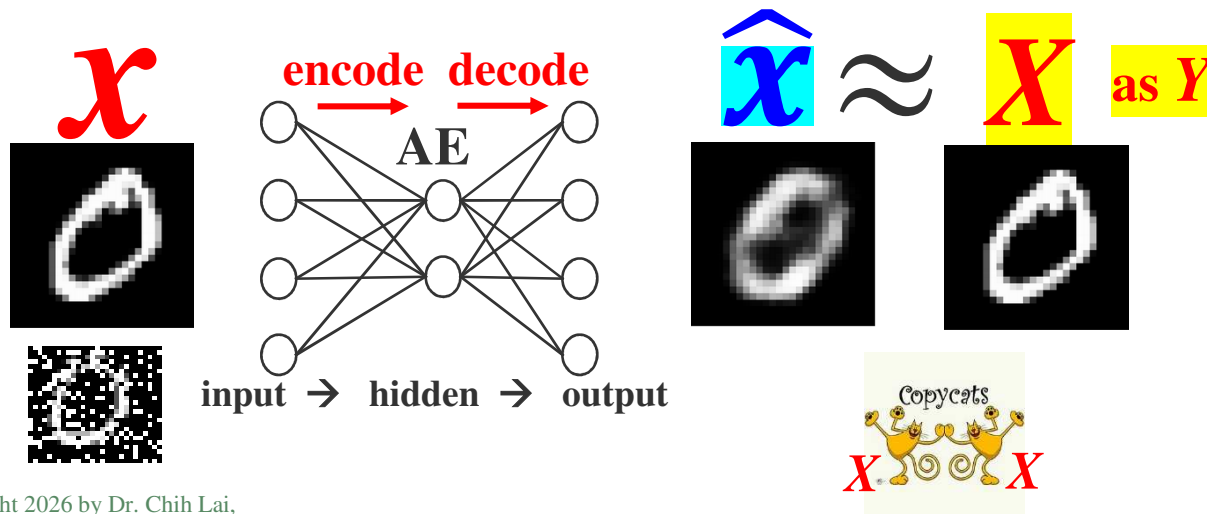


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- Quality Eval of Generation, Motion, Image Processing ➔ image light, resolution

Training in Regular NN vs. Autoencoder



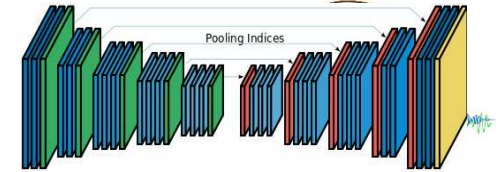
- Training autoencoder (AE), what is **Y**? → **semi-supervised** learning
- Just set $Y = X$, like copycat, BUT WHY??



1. Extract Z w/o Y
2. Unbalance classes
3. Denoising
4. Train deep NN w/o Y
5. Outlier detection



CNN Encoder + Decoder, Predict Next Video Frame

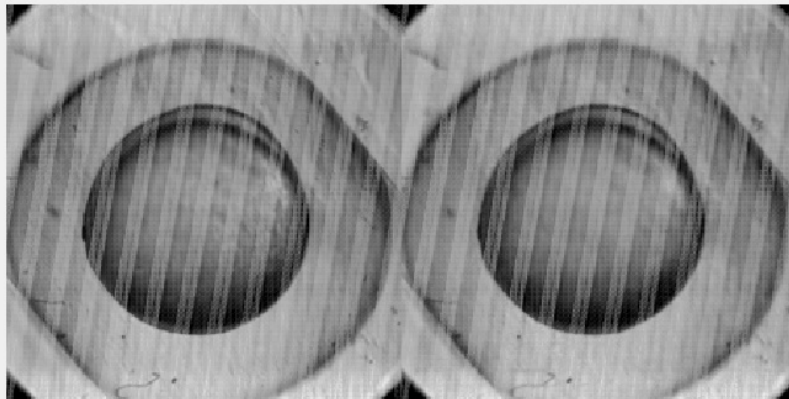


- Predicting Zebra Fish Embryo Development Over 8.5 Hours
 - Original 1-minute video captures 24-hour embryo development.

begin at 12-hour

Time up to 30

24-min

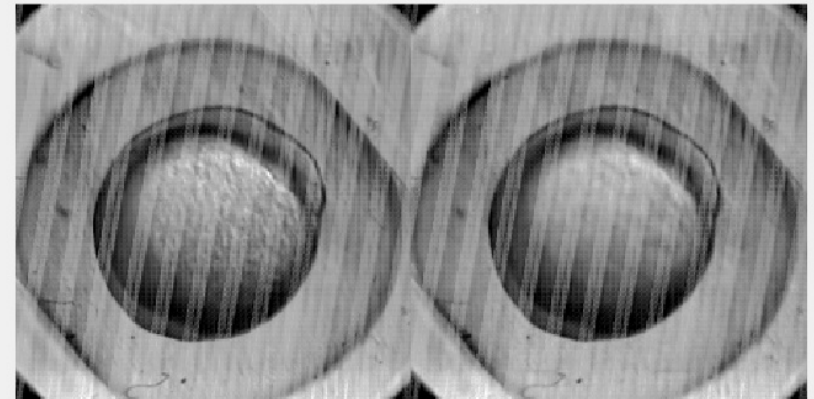


original

predicted

Time up to 210

168-min

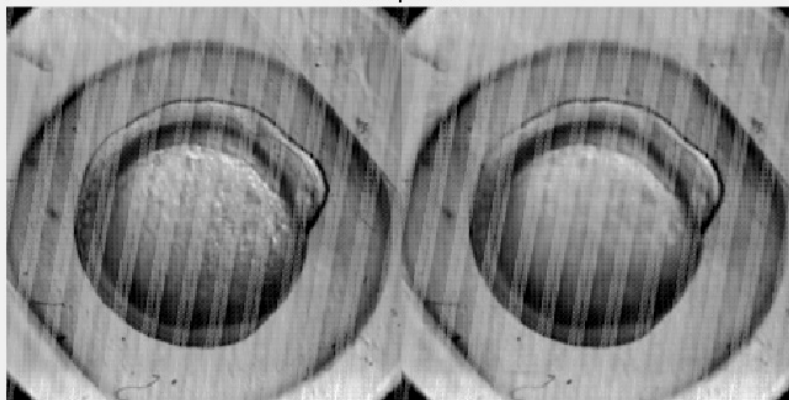


original

predicted

Time up to 360

288-min



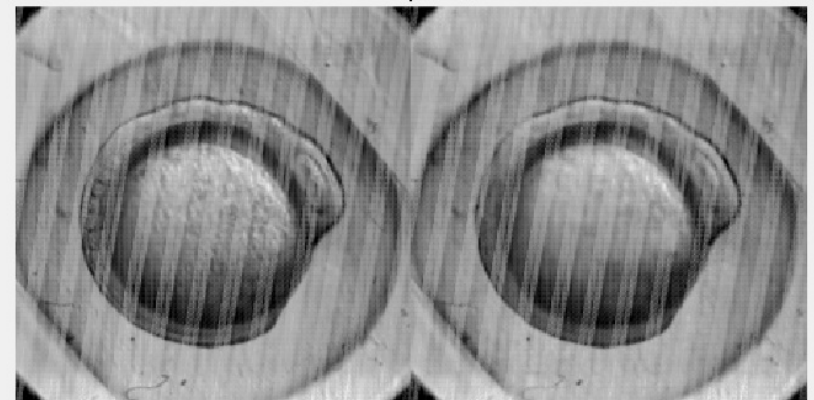
original

predicted

end at 20.2-hour

Time up to 460

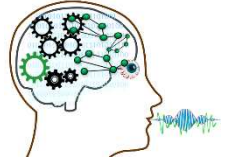
368-min



original

predicted

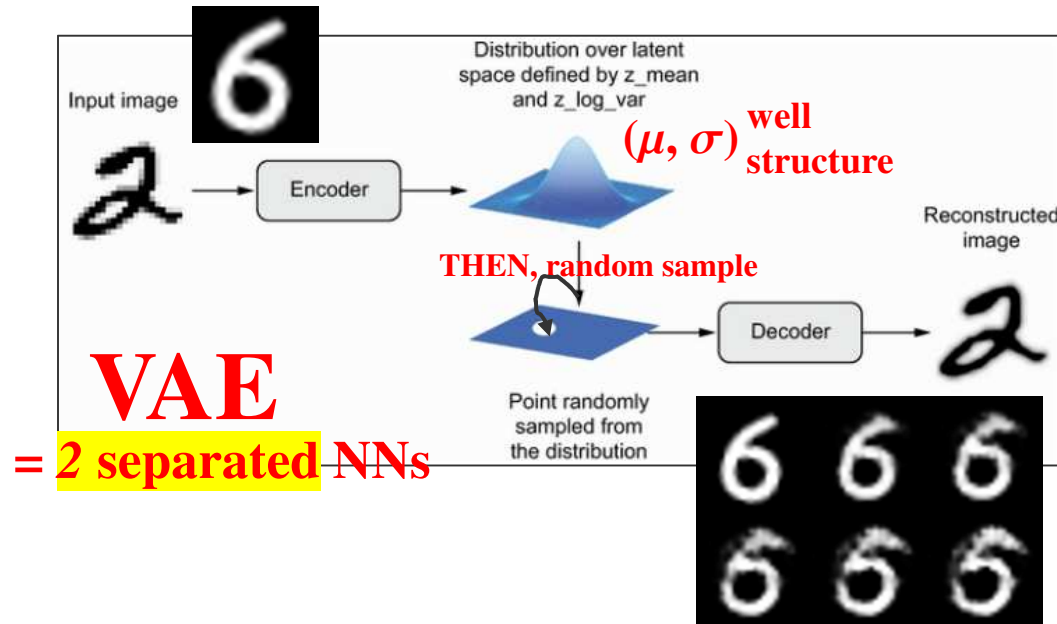
Outline



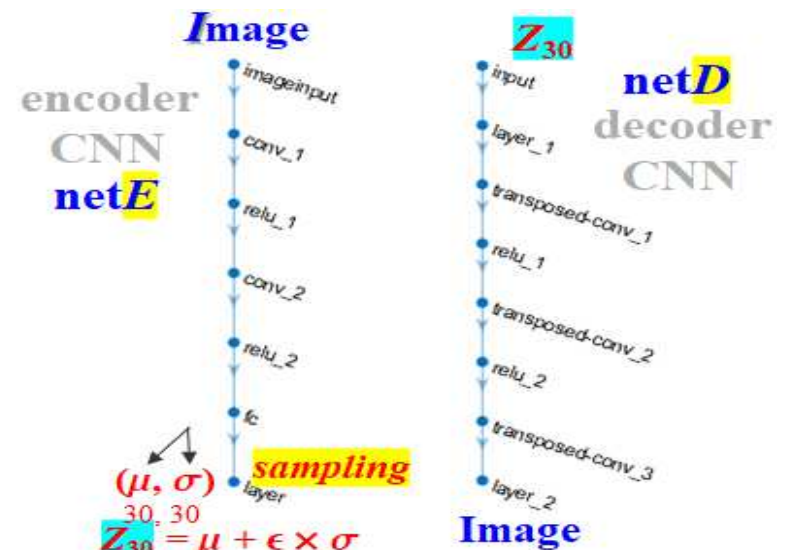
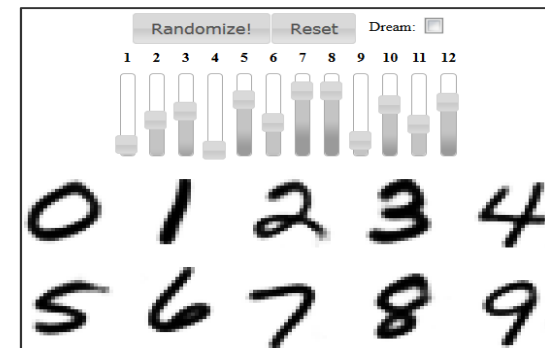
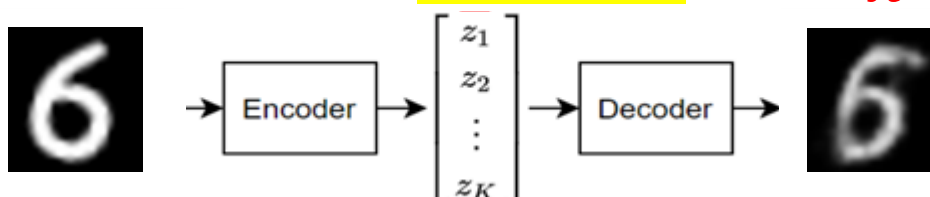
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Variational Auto-Encoder (VAE)

- VAE = early type of Generative AI
 - For generating images with variations



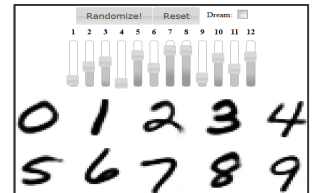
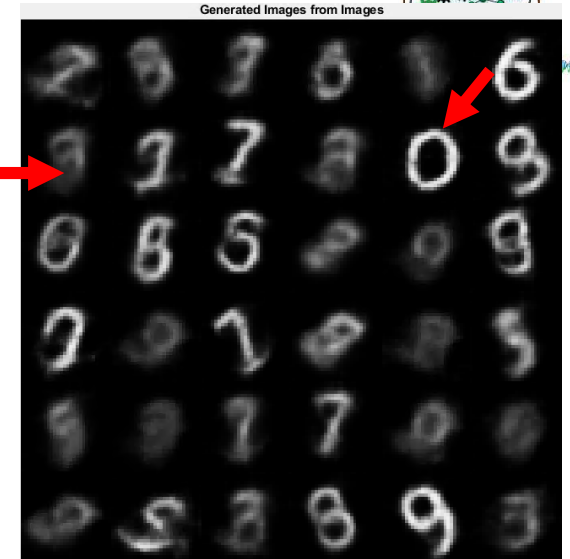
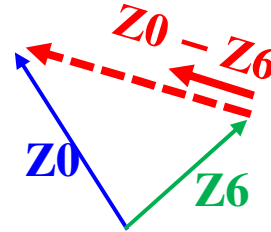
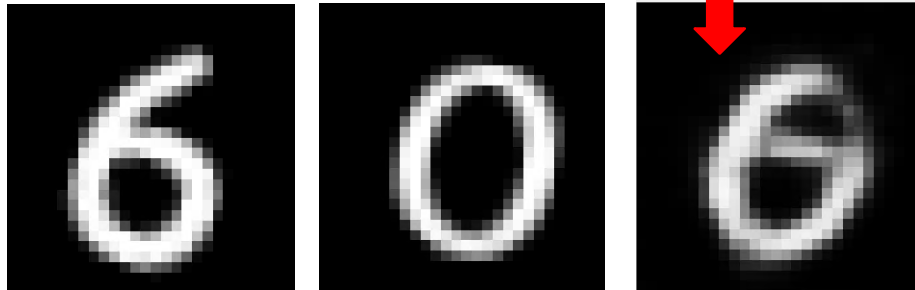
x **AE = 1 connected NN** $\hat{x}$



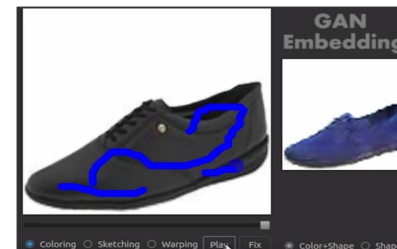


Walking One Z Toward Another Z

walking from digit 6 to digit 0



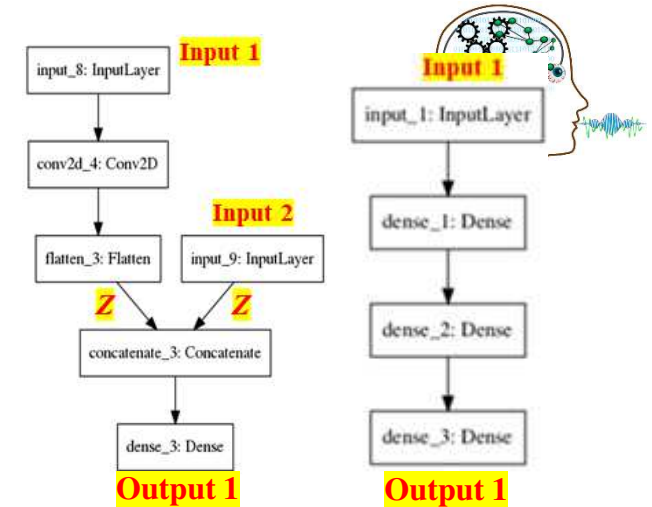
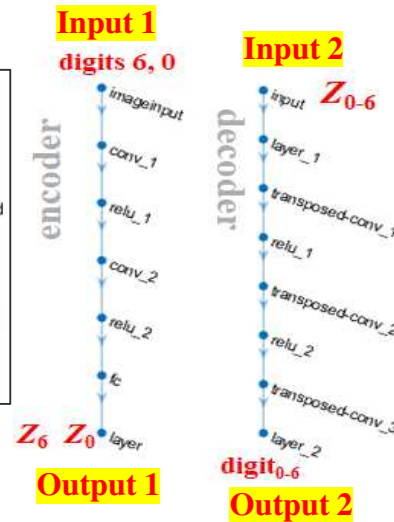
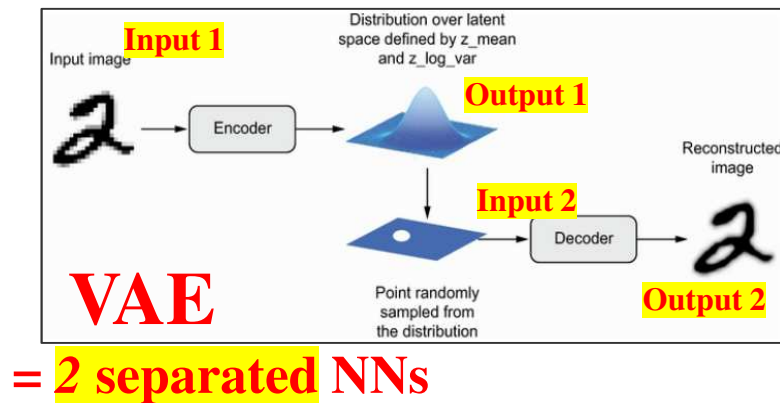
walking...



<https://www.youtube.com/watch?v=9c4z6YsBGQ0>

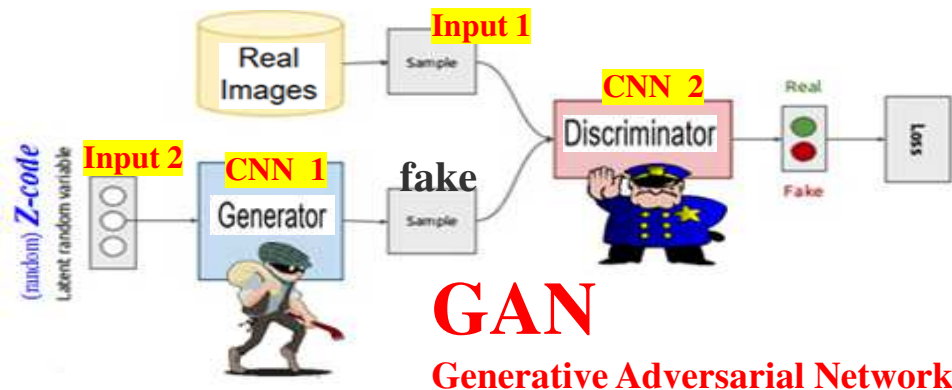
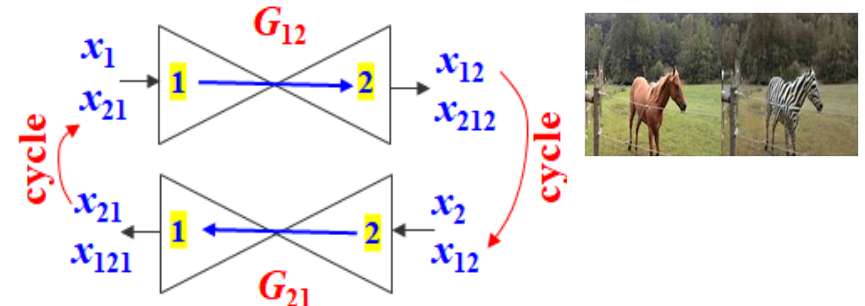
Special Network(s) Require Special Training

- 1 network with multiple inputs, **same loss**
- 2 networks with **different loss functions**

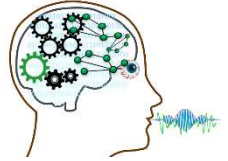


`model.fit(X, Y)`
`trainNetwork(X, Y)`

CycleGAN

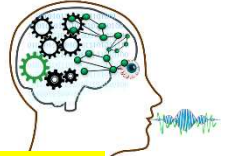


Outline



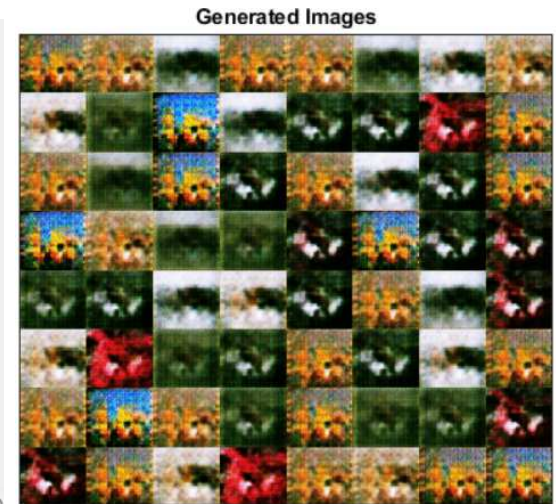
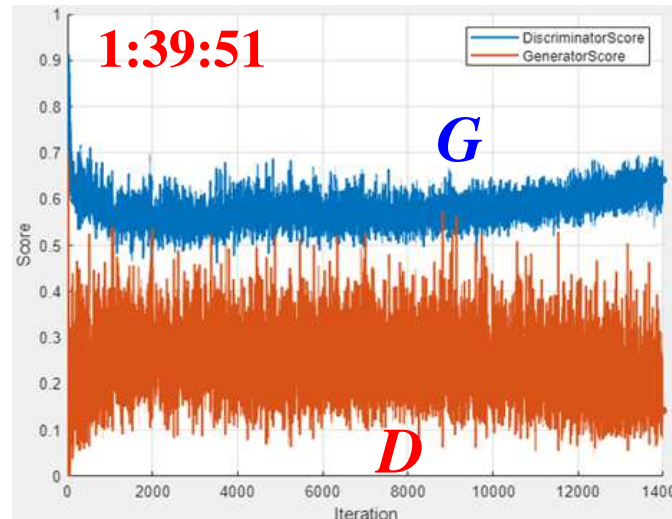
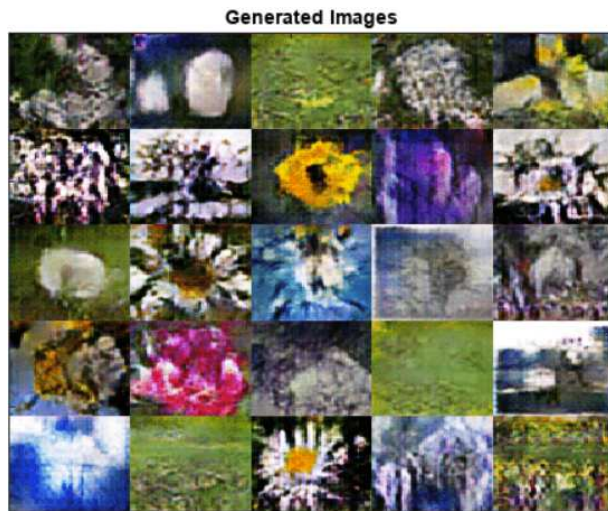
- **Z** / Visual Embedding
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- Quality Eval of Generation, Motion, Image Processing ➔ image light, resolution

GAN on Image Data

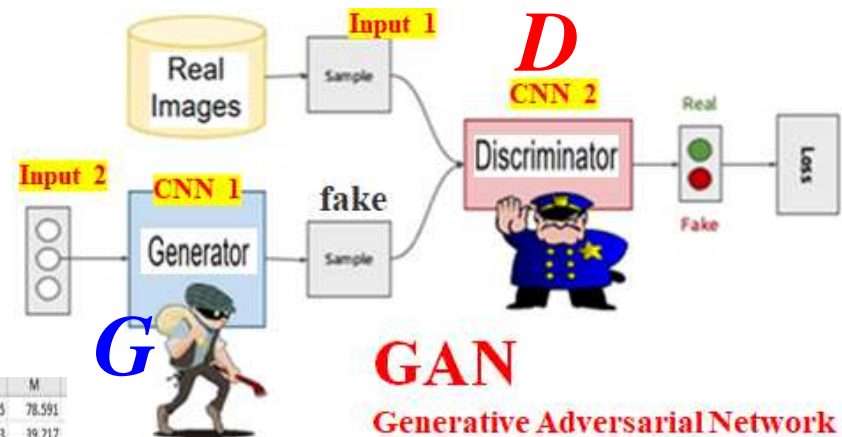
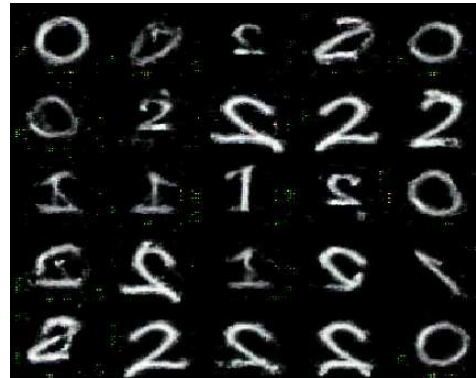
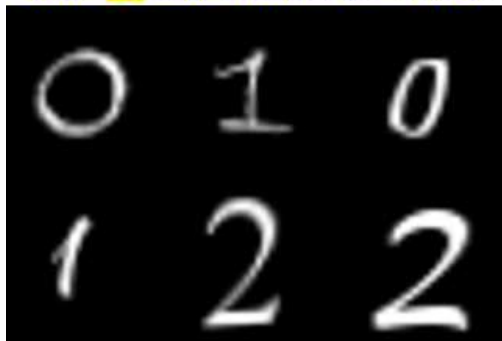


■ Generative Adversarial Network (GAN)

Mode Collapse



Train 30 Digit Images (of 0, 1, 2)



GAN for
numeric data
WHY???

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	222	31.18919	40.34234	35.57909	8.883917	0.968325	-80.1137	222	1	16.81247	0.816176	0.578125	78.591
2	73	29.49315	271.3973	15.5172	6.40749	0.910764	76.04295	73	1	9.640876	0.858824	0.608333	39.217
3	256	58.81641	289.9414	37.22601	9.863895	0.964256	85.32474	256	1	18.05407	0.752941	0.562637	89.111
4	126	71.02381	477.4127	13.11298	12.79067	0.220351	63.52348	126	1	12.66602	0.881119	0.646154	43.832
5	225	90.80889	541.9467	44.46311	7.858879	0.984256	-52.875	225	1	16.92569	0.728155	0.252525	90.072
6	233	91.40773	279.6223	40.03947	9.256891	0.972908	80.28062	233	1	17.22396	0.732704	0.524775	88.623
7	188	106.7118	187.2883	43.30114	6.184347	0.676231	-4.63487	188	1	18.14833	0.843106	0.718663	87.183

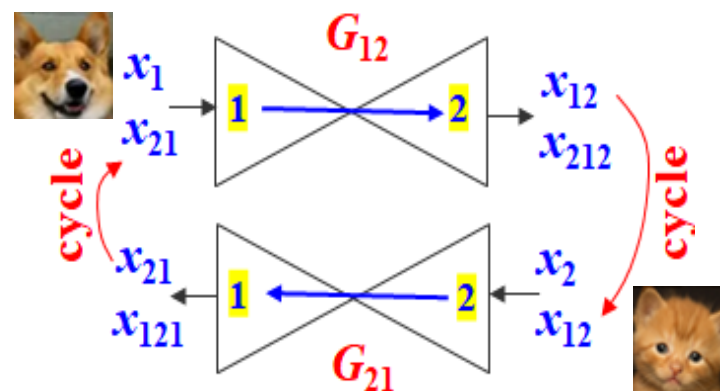
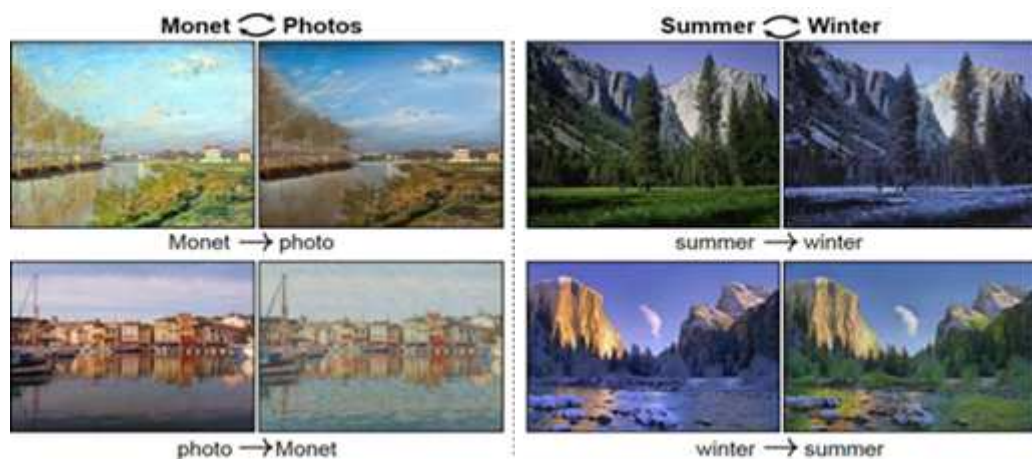


2017, <https://junyanz.github.io/CycleGAN/>

CycleGAN, UNIT GAN

(UNsupervised Image-to-image Translation)

- Across **2** Domains, **NO** Image Pairing Needs
- More serious applications???



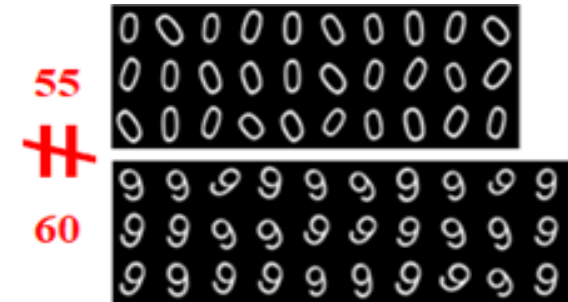
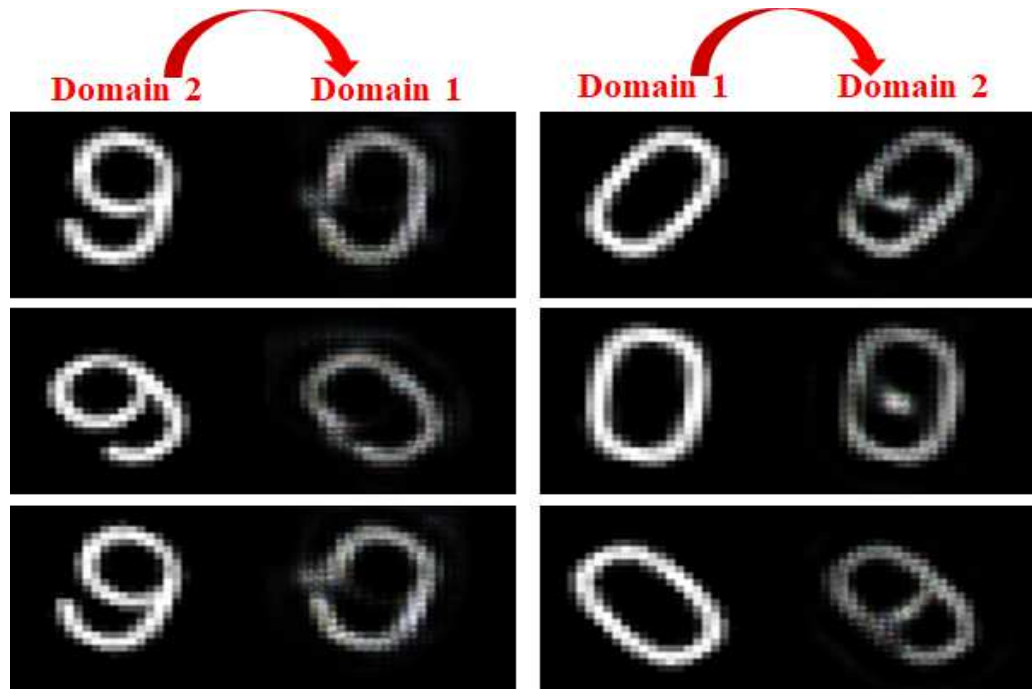
<https://rom1504.medium.com/image-embeddings-ed1b194d113e>



<https://junyanz.github.io/CycleGAN/>

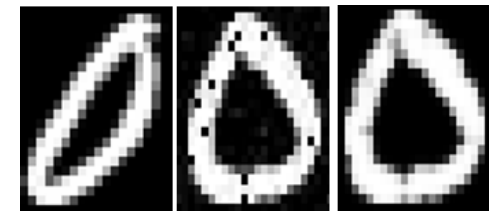
CycleGAN / UNITGAN Experiments

NO image pairing required

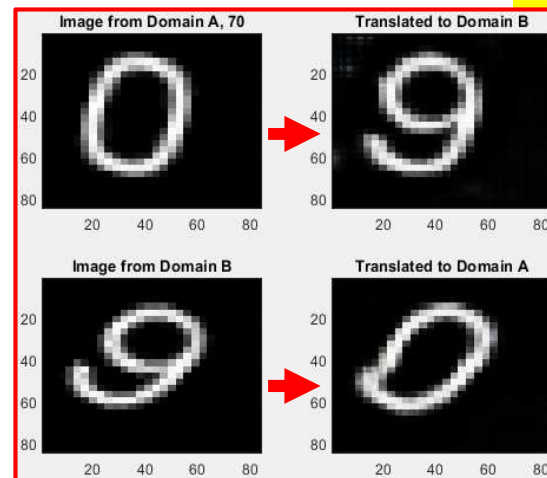
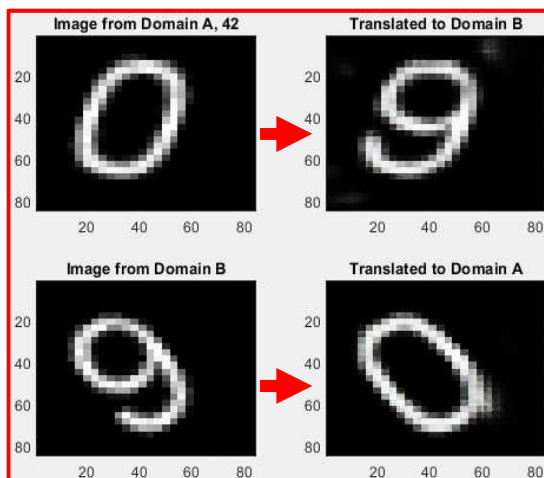


Cycle-GAN

denoising



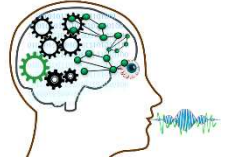
UNIT-GAN



CycleGAN for
numeric data
WHY???

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	222	31.18919	40.34234	35.57909	8.883917	0.968325	-80.1137	222	1	16.81247	0.816176	0.578125	78.591
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3	256	58.81641	289.9414	37.22601	9.863895	0.964256	85.32474	256	1	18.05407	0.732941	0.562637	89.111
4	126	71.02381	477.4127	13.11298	12.79067	0.220351	63.52348	126	1	12.66602	0.881119	0.646154	43.832
5	225	90.80889	541.9467	44.46311	7.858879	0.984256	-52.875	225	1	16.92569	0.728155	0.252525	90.072
6	233	91.40773	279.6223	40.03947	9.256891	0.972908	80.28062	233	1	17.22396	0.732704	0.524775	88.623
7	782	100.7118	187.7881	43.30114	6.184347	0.676011	-6.63687	782	1	10.148313	0.843106	0.738463	87.183

Outline

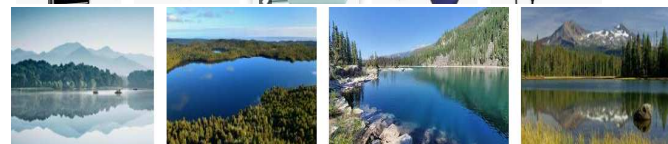
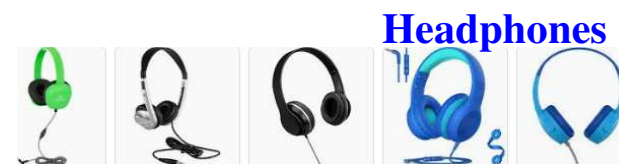


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We know AI can do many things... How about???

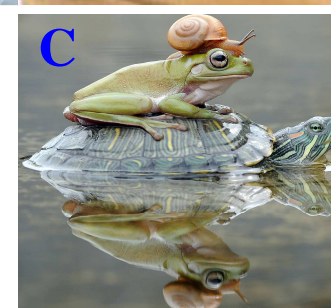


- Current AI can accurately classify 1000+ objects / classes *individually*



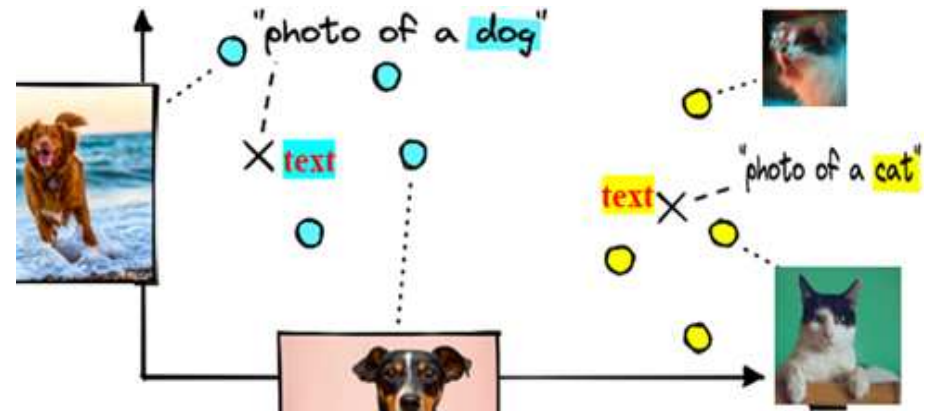
- But, how about *classifying* images **A, B, C?**
describing

- Current AI??
 - Cannot recognize combinations of known objects
 - i.e. **NO** compositional generalization / classification
 - How about unknown object?

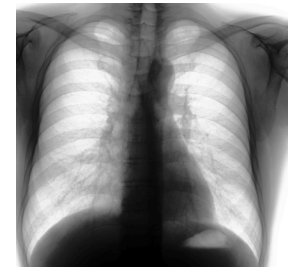
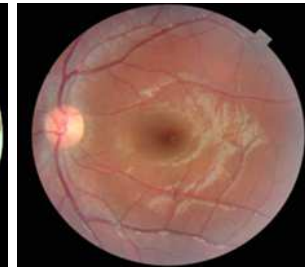


Association of Anything

Describe images?? Image captions??

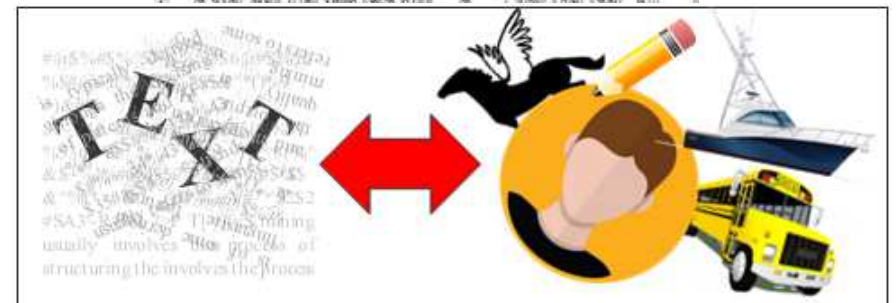


What diseases??



- Text
 - *Show me images w/ dog & cat &*
 - *Show me x-ray w/ disease A & B & C & ...*
- What is my point?? So far...
- Train model for each class for 1 info type
- Can we train *multi-modal model*?
- Can we do **0-shot** w/o keep re-training??
 - on new data of unseen classes of different types

1	222	20.0894	40.0424	20.5788	0.00907	0.00025	-0.0137	202	1	0.0000	0.00000	0.570125	70.591
2	79	20.4915	27.3973	15.0172	0.40749	0.00704	-0.00000	79	1	9.94007	0.050024	0.000013	59.217
3	164	20.0161	20.0161	27.7349	0.402085	0.00704	0.00000	164	1	10.0000	0.00000	0.000000	60.111



CLIP Predictions from Image Walking Results



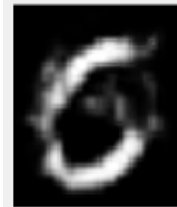
This plot accurately depicts the presence of digit 0. (0.99)
The number 0 can be found in this image. (0.99)
In this image, the occurrence of the digit 0 is evident. (0.99)



This image represents the digit zero. (0.84)
The digit 0 is clearly depicted in this image. (0.84)
The occurrence of the numeral 0 is clearly visible in this image. (0.84)



The digit 2 appears in this plot. (0.88)
This plot provides a clear visualization of digit 2. (0.88)
This plot clearly displays the digit two. (0.88)



The plot effectively communicates the presence of the digit 2. (0.83)
The plot features the numeral 2. (0.83)
The plot clearly indicates the presence of the numeral 2. (0.83)

Very like zero-shot classification

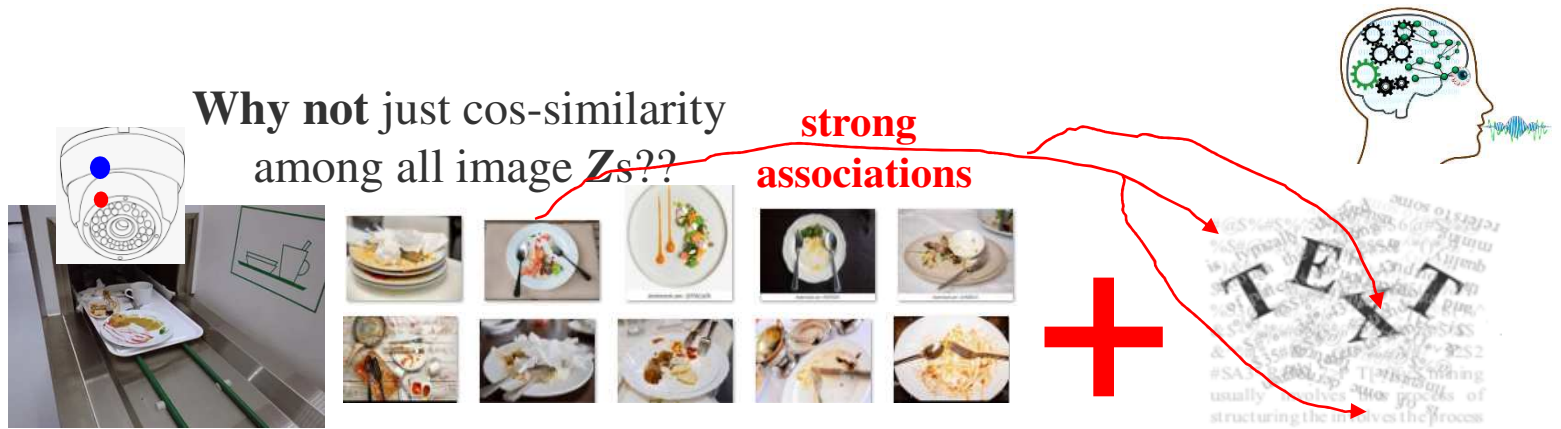


This image accurately illustrates the presence of the digit one. (0.95)
The number 1 can be found in this image. (0.95)
The plot effectively conveys the occurrence of the digit 1. (0.95)



The plot effectively communicates the presence of the digit 0. (0.94)
The digit 0 is clearly depicted in this image. (0.94)
The number 1 can be found in this image. (0.94)

Food Waste Application



Text Query Enter **Food Waste**

Select Image

Quit

Image 62, Vegetable Scraps, Wilted Lettuce, Chopped Vegetables, Leftover Condiments

Image 36, Salad Components: Leafy greens, Salad Components: Bell pepper, Pasta Dish. CosSim = 0.96164

Image 185, meat, vegetables, sauces. CosSim = 0.95465

Image 3, radishes, leafy greens, vibrant sauce. CosSim = 0.94071



Images + **TEXT**

Query
Image
62



<https://www.winnowsolutions.com/>

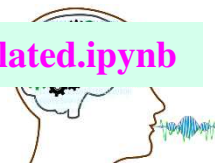
<https://www.leanpath.com/>

■ Image query using Text:

- “Vegetable Scraps, Wilted Lettuce, Chopped Vegetables, Condiments”

Characteristics CLIP vs. BLIP-2

VAI_Slide_BLIP2_ExeFlow_Simulated.ipynb

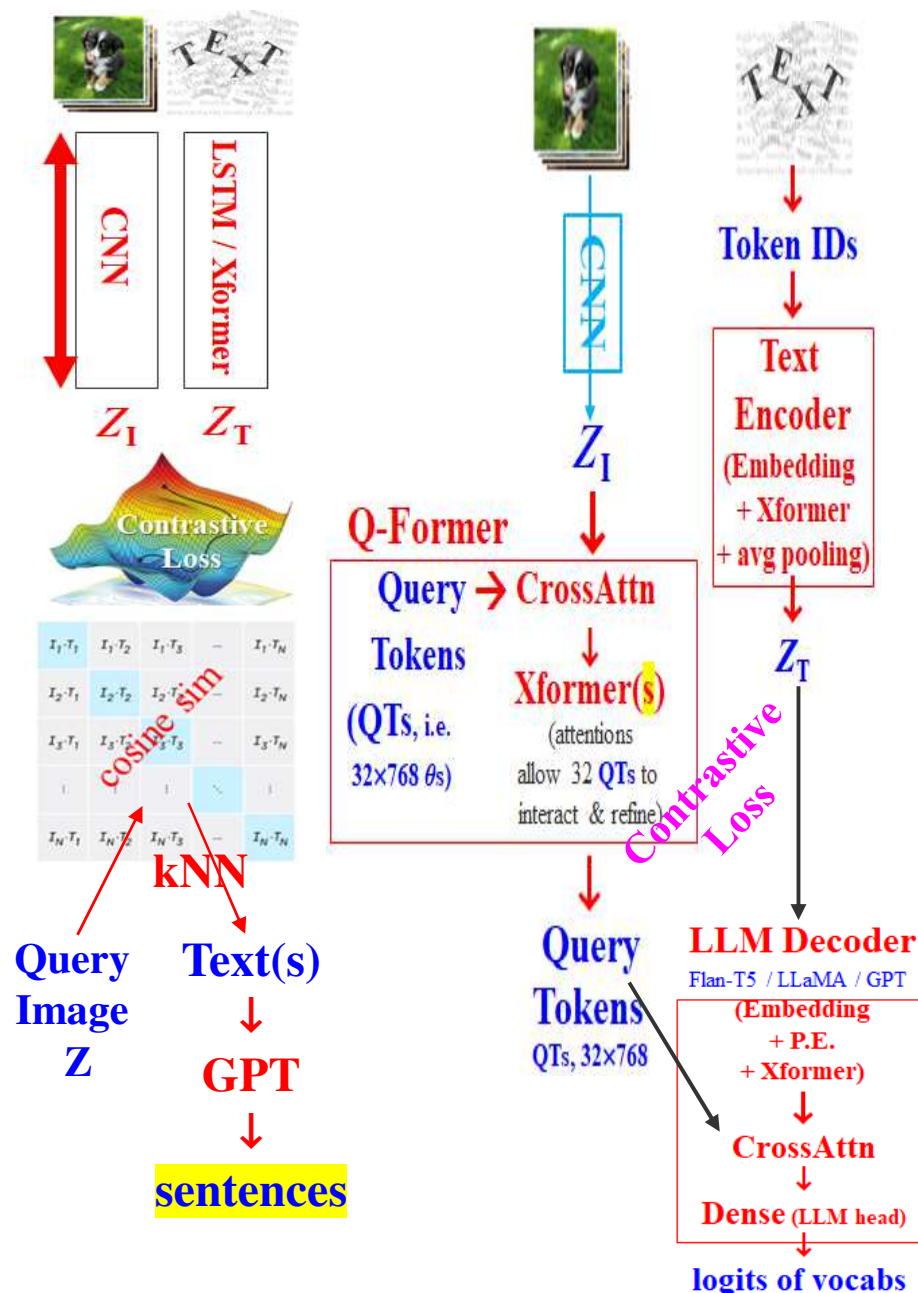


■ CLIP

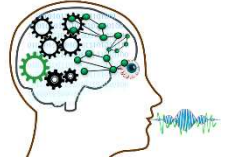
- Area: works with **any** pair of data **types**
- Good for **retrieval, classification, scoring**
- Example:
Describe: blood cell type: red, white, platelet.
Describe: x-ray conditions,
Describe: “traffic jam,” “street market,” “concert”

■ BLIP-2

- Major area **image** → **text** (not general multimodal mapping)
- Good for general description, **captioning, reasoning, Q&A**
- Example: Cars are lined up bumper to bumper on a crowded city street.
- A golden retriever is seen enjoying a brisk morning run in a park.....



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Knowledge Distillation (KD), Self-Distillation (SD) Self-Supervised Learning (SSL)



1. Knowledge Distillation

CVPR, ICLR

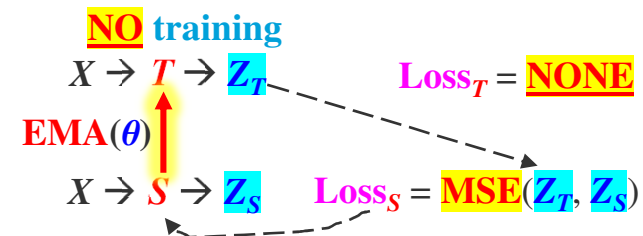
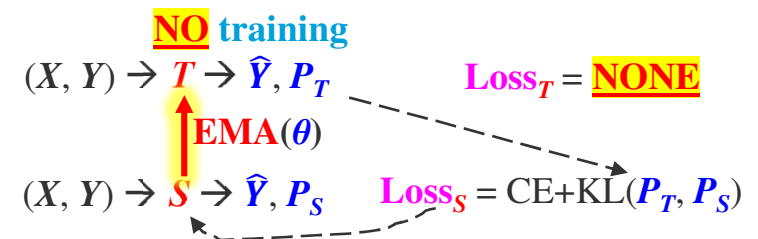
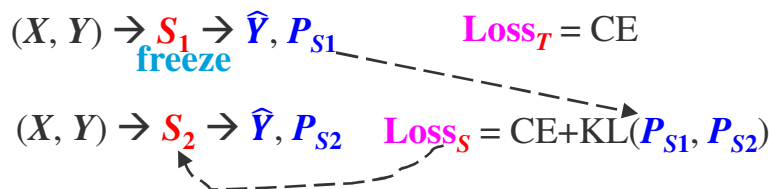
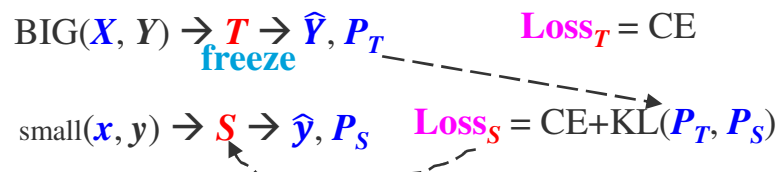
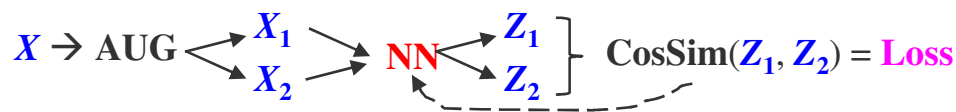
2. Self Distillation / BAN (Born-Again)

ICML 2018

3. Self Distillation w supervised / unsupervised EMA (Exponential Moving Average)

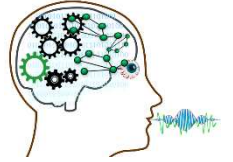
4. Self-Supervised Learning / SSL,

- **SimCLR, BYOL** By Google Research '20, Google DeepMind '20
- DINO, **DINO v2** By Meta '21, Apr. Meta '23
- **DINO / v2** = SSL + **Supervised / Unsupervised** EMA teacher + feature matching + self-distillation



2/12/2026

Outline

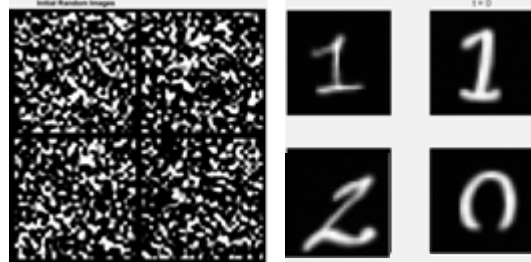
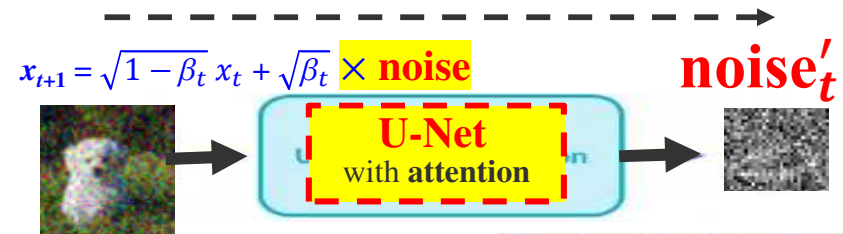
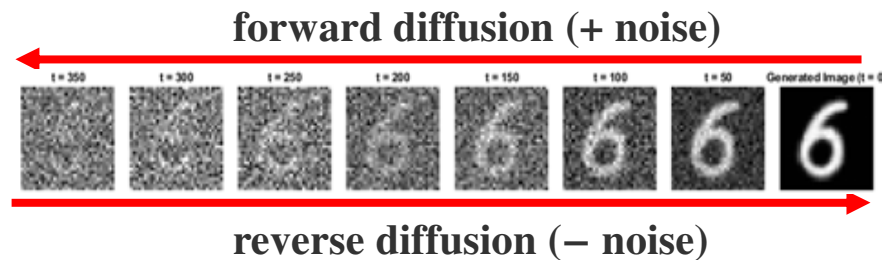


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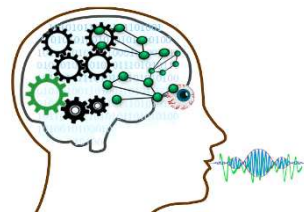
Denoising Diffusion Probabilistic Model (DDPM)



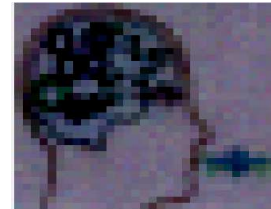
- Can CNN learn the pattern of “*noise diffusion*” so it can do reversed process?



DiffuseDigitDemo.avi



Class_Logo_32x32_Version_2.mp4



- But, unpredictable / unstable outputs

- Stable diffusion is a free alternative of OpenAI's DALL-E, trained on **10⁹** (billion) images

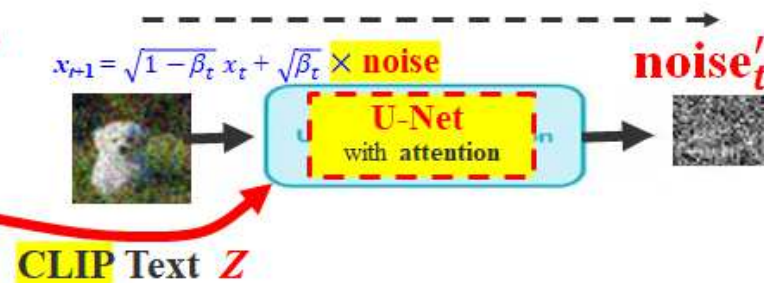


<https://stablediffusion.fr/demo>

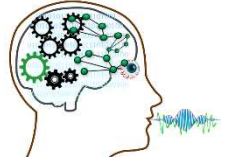
How to
Stable Diffusion?

TEXT prompt →

Dog ridding on
the back of a hippo



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Image Generation, Iteratively Update Images, **NOT** CNN

■ DeepDream



Input Image



Pyramid scale 1:1



1:5



1:8

■ Neural Style Transfer (NST)

→ Content target



→ Style reference



+

=



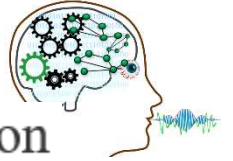
<https://link.springer.com/article/10.1007/s11042-018-6929-x>



Apps more than artistic / entertainment??

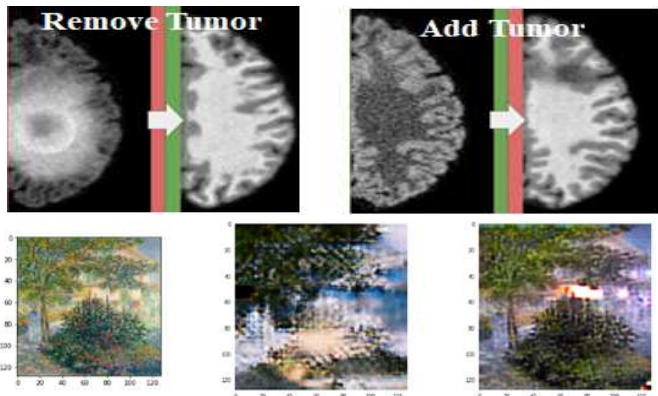
As **image augmentation**

Outline



- Z / Visual Embedding
- Object Allocations by Pixel Classification
- “COLORs” to VAI as “Words” to Conv. AI
- Auto-Encoder / Decoder, Rebuild Input with Few Y labels, Denoising
- Image Generation or Numeric Data Generation Special Networks & Loss
 - Variational Auto-Encoder (VAE), CGAN, regularized WGAN, mode collapse
 - CycleGAN, UNIT GAN, Image Translation 2 Domains, Style Transfer / Stable Diffusion
- Multi-Modal Model, Images + TEXT + Audio CLIP, BLIP
- Faster / Better Training w/ Knowledge Distillation, Self-Supervised Learning **NO** Y

➔ Quality Eval of Generation



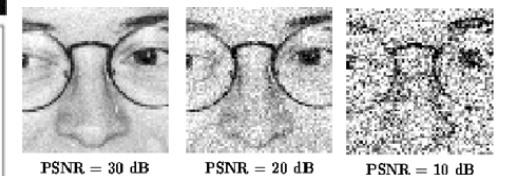
input

noisy

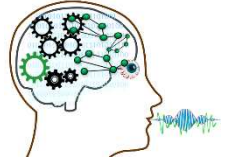
de-noisy

NIQE = 18.8722

NIQE,	L2,	MSE,	SSIM,	PSNR,	FID
18.8701	4.32298	0.0182501	0.456763	17.3873	8.44347
NIQE,	L2,	MSE,	SSIM,	PSNR,	FID
18.8727	2.21062	0.0047723	0.917965	23.2127	1.22555



Outline



- Z / Visual Embedding
- Object Allocations by Pixel Classification
- “**COLORS**” to VAI as “Words” to Conv. AI
- Auto-Encoder / Decoder, Rebuild Input with Few *Y* labels, Denoising
- Image Generation or Numeric Data Generation Special Networks & Loss
 - Variational Auto-Encoder (VAE)
 - **G**enerative **A**dversarial **N**etwork (GAN), **C**GAN, regularized **W**GAN, mode collapse
 - *CycleGAN*, *UNIT GAN*, **Image Translation** in 2 Domains
 - Style Transfer / Deepdream / Stable **Diffusion**
- Multi-Modal Model, Images + TEXT + Audio
 - **C**ontrastive **L**anguage-**I**mage **P**re-training (**CLIP**), Associate *Anything*, **BLIP** captioning
- Faster / Better Training w/ Knowledge Distillation, Self-Supervised Learning **NO** *Y*
- Quality Eval of Generation, Motion, Image Processing → image light, resolution

→ Summary

Quick 766 Topic Summary



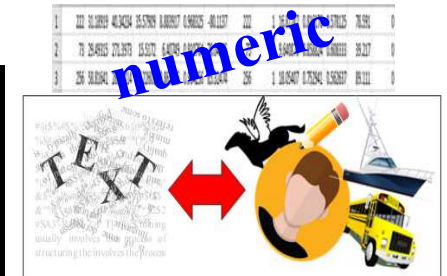
- Why care images? **80%** of 766 topics have numeric dataset examples (& homework)

- **AutoEncoder**: can still work under...

- **NO** Y, very **small** dataset, very **deep** NN, outlier detect, denoise

- **CLIP**: learn *association* between **ANY** datasets **images & text**

- **Multi-Modal Model**, **0-shot** prediction



- **GAN** (**G**enerative **A**dversarial **N**etwork)

- **Cycle-GAN**, **UNIT-GAN**

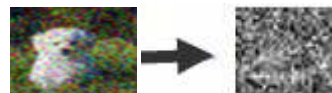


- **Object allocation**

NOT whole image
classification



- **Knowledge Distillation, Neural Style Transfer, Diffusion**



walking..  Z



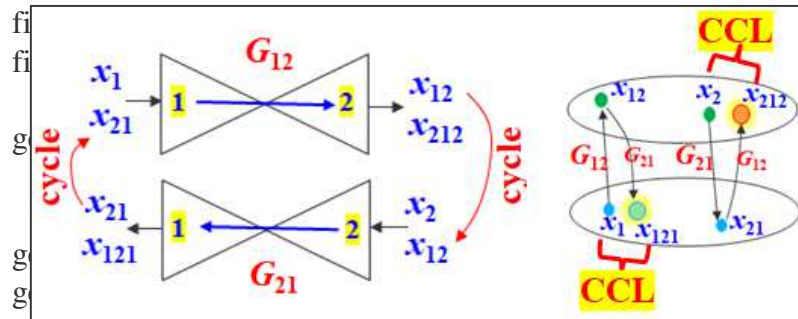
Real Program & Loss Functions



```
cycleImageA2B2A = forward(genB2A,imageBGenerated);
cycleImageB2A2B = forward(genA2B,imageAGenerated);
```

```
cycleLossA2B2A = cycleConsistencyLoss(imageA,cycleImageA2B2A,lambda);
cycleLossB2A2B = cycleConsistencyLoss(imageB,cycleImageB2A2B,lambda);
```

```
identityImageA = forward(genB2A,imageA);
identityImageB = forward(genA2B,imageB);
```



```
discAScore = 0.5*mean(sigmoid(predGeneratedA,"all")) + ...
discBScore = 0.5*mean(sigmoid(predGeneratedB,"all")) + ...
scores = {genB2AScore, genA2BScore, discAScore, discBScore};
genA2BGrad = dlgradient(genLossTotal,genA2B.Learnables,RetainData=true);
genB2AGrad = dlgradient(genLossTotal,genB2A.Learnables,RetainData=true);
```

Partial Pseudo Code

Full Program

```
% Translate images from one domain to another: low-dose to high-dose
% vice versa
[ImageAGenerated,genB2AState] = forward(genB2A,imageB);
[ImageBGenerated,genA2BState] = forward(genA2B,imageA);
% Calculate predictions for real images in each domain by the
% corresponding discriminator networks
predRealA = forward(discA,imageA);
predRealB = forward(discB,imageB);
% Calculate predictions for generated images in each domain by the
% corresponding discriminator networks
predGeneratedA = forward(discA,imageAGenerated);
predGeneratedB = forward(discB,imageBGenerated);
% Calculate discriminator losses for real images
lossRealA = lossReal(predRealA);
lossRealB = lossReal(predRealB);
% Calculate discriminator losses for generated images
lossGeneratedA = lossGenerated(predGeneratedA);
lossGeneratedB = lossGenerated(predGeneratedB);
% Calculate total discriminator loss for each discriminator network
discALossTotal = 0.5*(discALossReal + discALossGenerated);
discBLossTotal = 0.5*(discBLossReal + discBLossGenerated);
% Calculate generator loss for generated images
genLossB2A = lossReal(predGeneratedA);
genLossA2B = lossReal(predGeneratedB);
% Complete the round-trip (cycle consistency) outputs by applying
% generator to each generated image to get the images in the
% original domains
cycleImageA2B2A = forward(genB2A,imageBGenerated);
cycleImageB2A2B = forward(genA2B,imageAGenerated);
% Calculate cycle consistency loss between real and generated images
cycleLossA2B2A = cycleConsistencyLoss(imageA,cycleImageA2B2A,lambda);
cycleLossB2A2B = cycleConsistencyLoss(imageB,cycleImageB2A2B,lambda);
% Calculate identity outputs
identityImageA = forward(genB2A,imageA);
identityImageB = forward(genA2B,imageB);
% Calculate fidelity loss (SSIM) between the identity outputs
fidelityLossA = mean(1-multissim(identityImageA,imageA),"all");
fidelityLossB = mean(1-multissim(identityImageB,imageB),"all");
% Calculate total generator loss
genLossTotal = genLossB2A + cycleLossB2A2B + fidelityLossA + fidelityLossB;
% Calculate scores of generators
genB2AScore = mean(sigmoid(predGeneratedA),"all");
genA2BScore = mean(sigmoid(predGeneratedB),"all");
% Calculate scores of discriminators
discAScore = 0.5*mean(sigmoid(predGeneratedA),"all") + ...
0.5*mean(1-sigmoid(predGeneratedA),"all");
discBScore = 0.5*mean(sigmoid(predGeneratedB),"all") + ...
0.5*mean(1-sigmoid(predGeneratedB),"all");
% Combine scores into cell array
scores = {genB2AScore, genA2BScore, discAScore, discBScore};
% Calculate gradients of generators
genA2BGrad = dlgradient(genLossTotal,genA2B.Learnables,RetainData=true);
genB2AGrad = dlgradient(genLossTotal,genB2A.Learnables,RetainData=true);
% Calculate gradients of discriminators
discAGrad = dlgradient(discALossTotal,discA.Learnables,RetainData=true);
discBGrad = dlgradient(discBLossTotal,discB.Learnables,RetainData=true);
% Return mini-batch of images transforming A into B
imagesOutAAndBGenerated = {imageA,imageBGenerated};
% Return mini-batch of images transforming B into A
imagesOutBAndAGenerated = {imageB,imageAGenerated};
```

<https://www.mathworks.com/help/deeplearning/ug/unsupervised-medical-image-denoising-using-cycleGAN.html>

Why VAI Sounds Like More Complicated Than LLM???



Tell me more about Master of AI

<https://soe.tommiebot.com/>

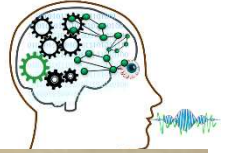
Tell me more about Master of AI in Data Science

Feature	Text	Images
Basic Unit	Discrete tokens (words/characters).	Millions of pixels (continuous/infinite).
Predictability	High: Constrained by grammar and logic (e.g., "I went to the store to... buy ")	Low: The next pixel or patch could be almost anything
Interpretation	Clear: Semantics and meanings are relatively fixed	Ambiguous: One object / scene has endless variations (light, angle, shadows)
Learning Objective	Stabilized: Settled on Xformers and next-token prediction w/ single entropy loss	Pre-paradigm: Require many different architectures w/ many different loss functions .
Complexity	Lower: Repeated patterns	Extremely High: 12B+ possible values in a single HD image, 48M pixels per image * 255 = 12B
Consistency	Reproducible: Mostly a matter of compute, data, and money	Variable: Hard to re-use; highly sensitive to noise and blur
Core Analogy	Fixed instructions in Lego or recipe	Sensitive, variable, and chaotic in Bucket of Sand or sand on floor

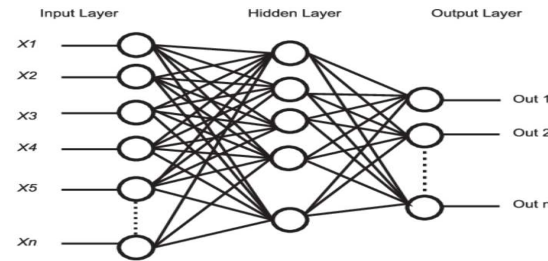


- One object / scene can appear under endless variations: *lighting, angle, occlusion, blur, background clutter, shadows, camera settings, environment*, etc. Most images & their semantics are ambiguous.

SEIS 764 Review Topics, and **MORE...**



- **Z-code**
- Augmentation
- Convolution
- Residual connection / purposes



- **Functional API**
- **NN training**
 - We will use multiple inputs (& outputs)

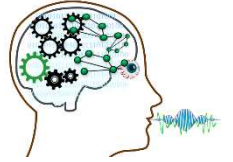
```
model = Sequential()
model.add(Dense(5, input_dim=8, activation='relu'))
model.add(Dense(3, activation='relu'))
model.add(Dense(2, activation='softmax'))
model.compile(loss='binary_crossentropy')
```

- **Training loop**
- **Loss / cross entropy / MSE**
- **Gradient descent learning**



```
options = trainingOptions('sgdm', ...
    'MaxEpochs', 50, 'MiniBatchSize', 100);
net = trainNetwork(ZX, Y, layers, options);
Yh = predict(net, ZX);
```

```
model.compile(loss='binary_crossentropy', optimizer='adam')
model.fit(X, Y_train, epochs=50, batch_size=4, verbose=0)
```

Appendix

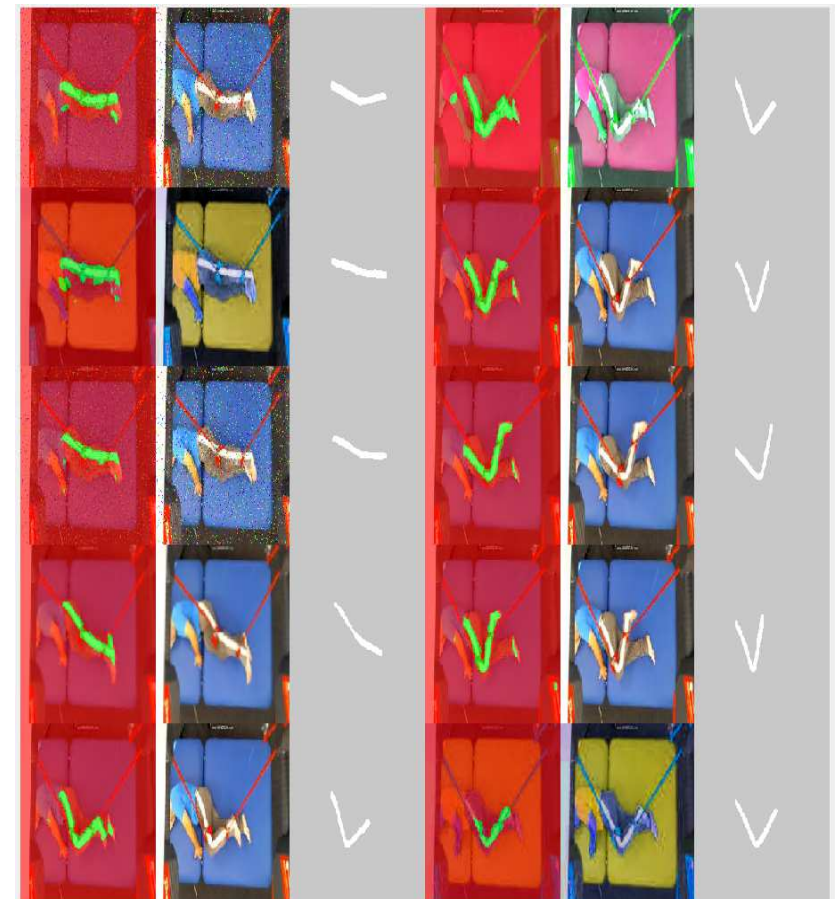
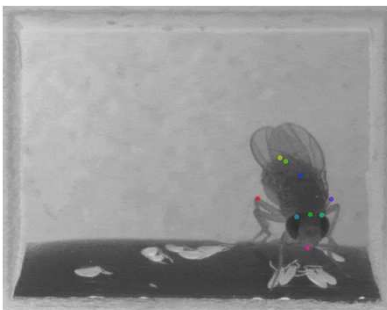
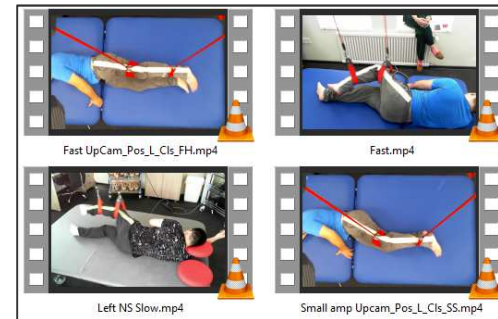
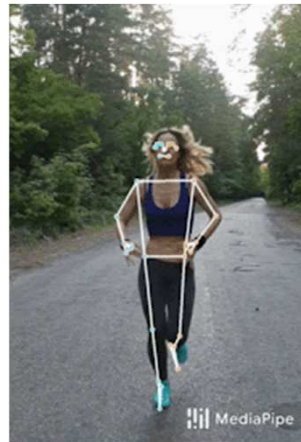
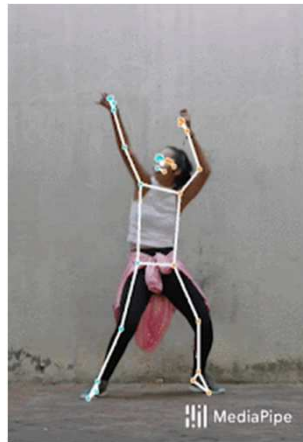
Motion & Examples

Pose Estimation

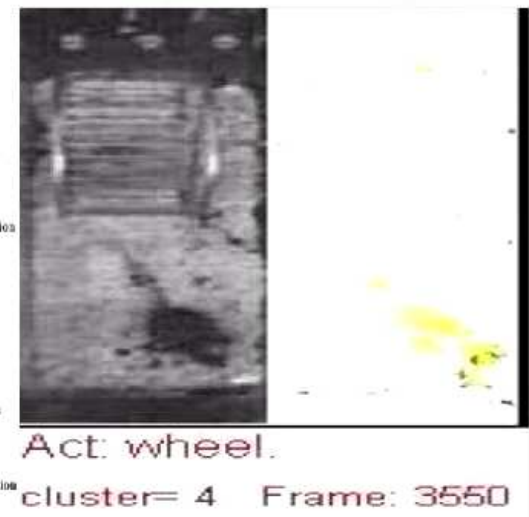
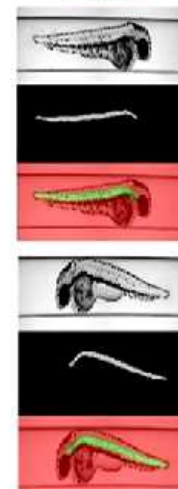
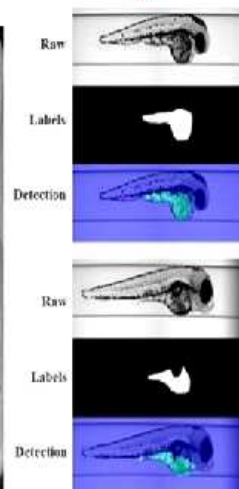
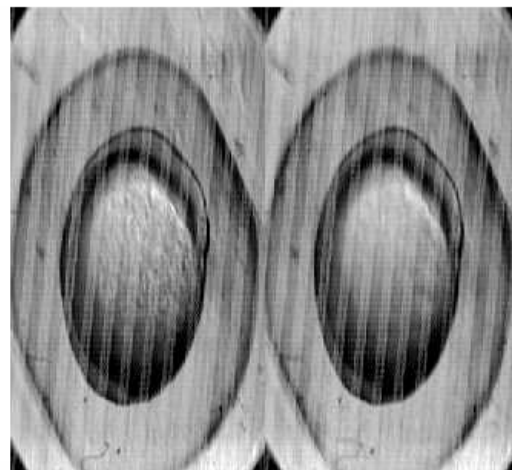
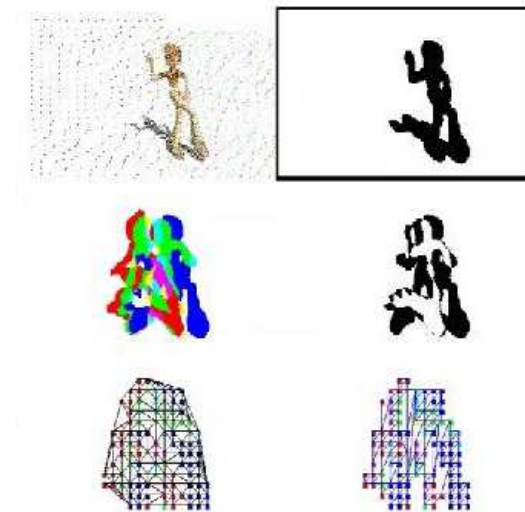
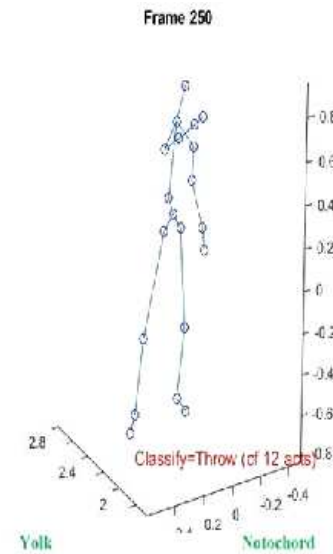
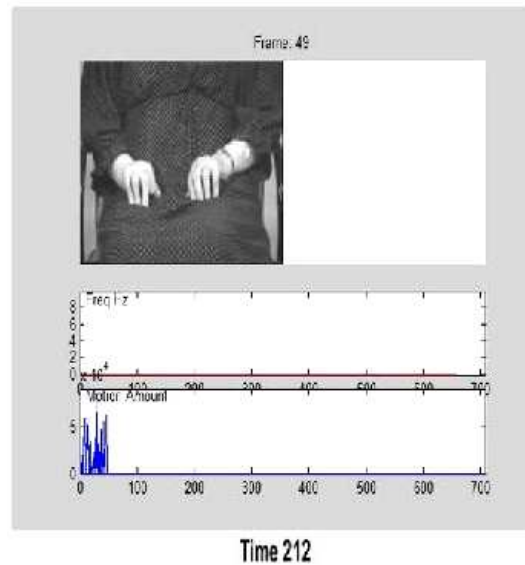
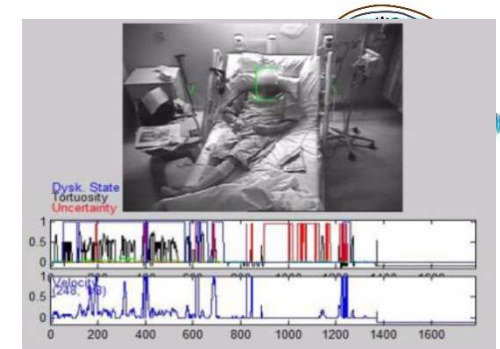


2020 with
Mayo Clinic

PoseNet
OpenPose



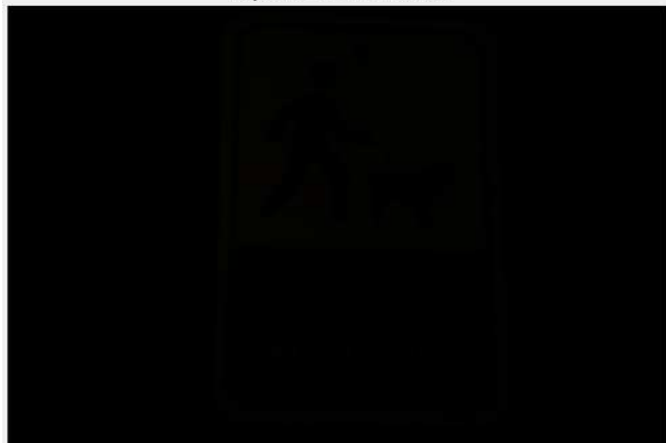
More Motion Detection Results



Recover Low-Light Images Using Deep Learning



Short Exposure Test Image s
Exposure Time = 0.04 s



Input

**Ground
Truth**

Long Exposure Target Image s
Exposure Time = 10 s



Prediction Time

Good CPU = 150 seconds

GPU = 1.9 seconds

Low-Light Recovery Network Prediction



**Recover
deep learn**

**Recover
tradition
method**

Histogram Equalization



<https://www.mathworks.com/help/images/learning-to-see-in-the-dark.html>

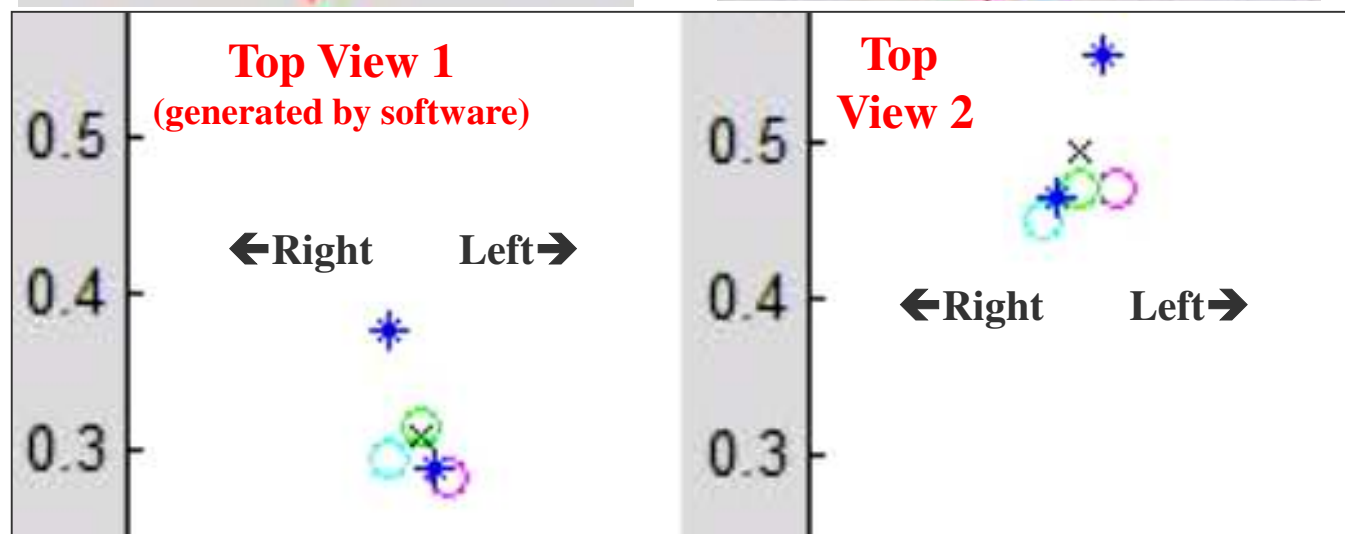
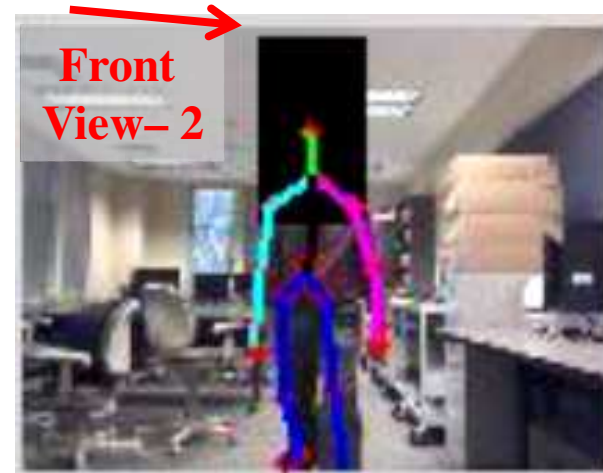
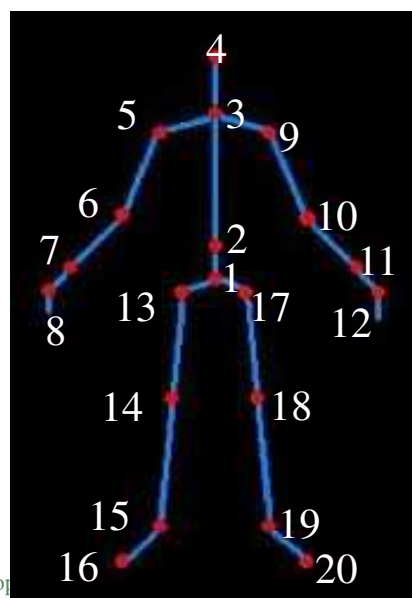
Trace 3-D Body Skeleton & Study Walking Gait with Microsoft Kinect



Use skeleton info. to trace & mask face to protect patient privacy

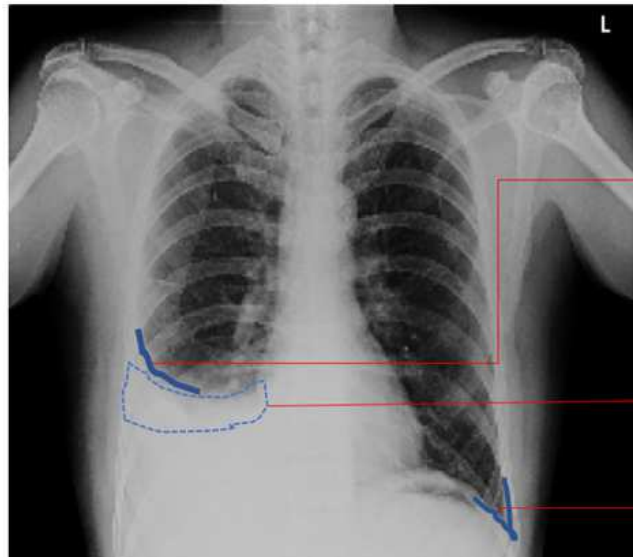
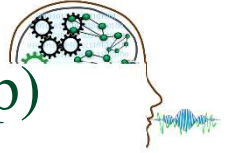


**Skeleton information
generated from
Kinect**



Legend: { ○ = Head × = Body Trunk * = Legs
○ = Left Shoulder ○ = Right Shoulder

Import Parts of Images? Occlusion Sensitive Map (OS-Map)



Blunt CP Angle

Accumulated Fluid

Sharp CP Angle

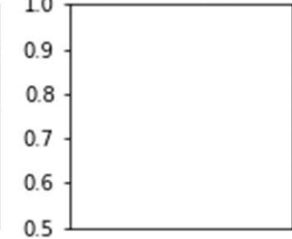
Original Image



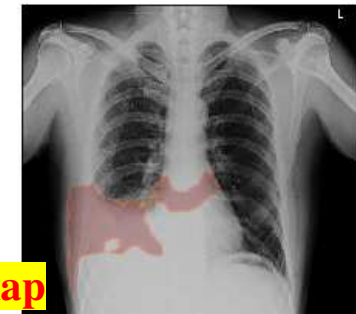
Perturbed Image



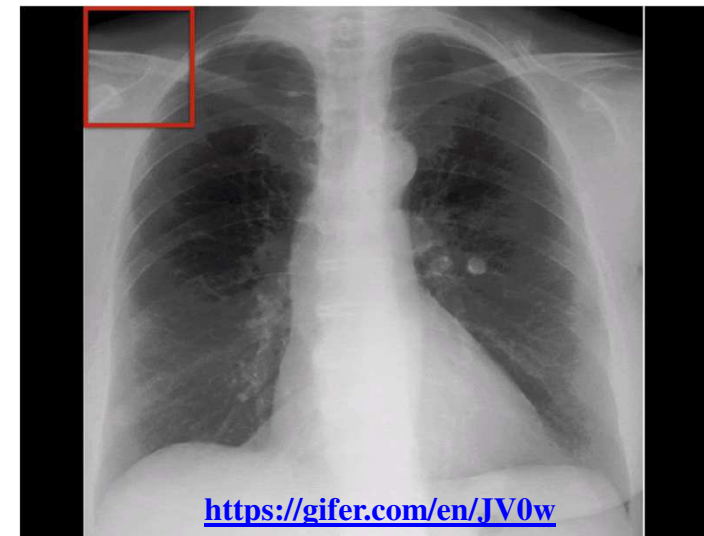
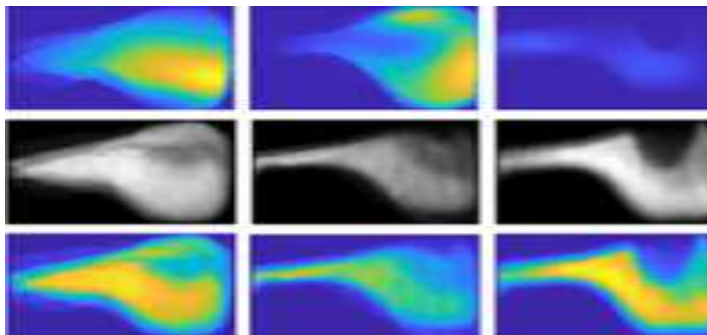
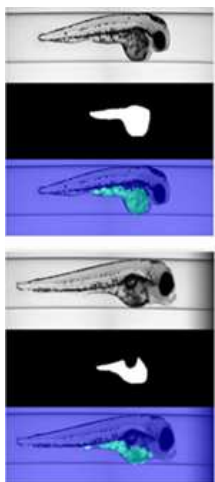
Probability plot



heat map

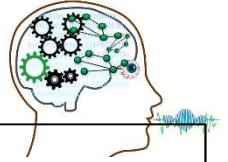


https://blog.qure.ai/notes/visualizing_deep_learning



<https://gifer.com/en/JV0w>

766 Topics from Different Angles



- Object Identification & more
- AutoEncoder (AE)
- Variational AE / VAE
- GAN-1: GAN, C-GAN, W-GAN
- GAN-2: Cycle-GAN, UNIT-GAN
- (Stable) Diffusion, Style Xfer, DeepDream
- CLIP / Classification

Grouping by Characteristics

- Object Identification & more
- AutoEncoder (AE), Variational AE (VAE)
- CLIP / Classification
- GAN-2: Cycle-GAN, UNIT-GAN
- Style Xfer
- GAN-1: GAN, C-GAN, W-GAN
- (Stable) Diffusion, DeepDream

Applicability

- Basic image processing, colors

- Object Identification & more

- AutoEncoder (AE)

- Variational AE / VAE

- **CLIP** (**C**ontrastive **L**anguage-**I**mage **P**re-training)

- GAN + **Quality** measures

- Cycle-GAN, UNIT-GAN

- C-GAN, W-GAN

- Style Xfer

- (Stable) Diffusion, DeepDream

- DNN for Image Processing

- Morphological Ops in traditional Image Processing

Discussion Order



model.**fit**(X, Y)
trainNetwork(X, Y)